

Review

A Narrative Review on Smart Sensors and IoT Solutions for Sustainable Agriculture and Aquaculture Practices

Liwei Liu ¹ , Winton Cheng ² and Hsin-Wei Kuo ^{2,3,*}

¹ Exhibition Division, National Science and Technology Museum, Kaohsiung 807412, Taiwan; liweiliu@mail.nstm.gov.tw

² Department of Aquaculture, National Pingtung University of Science and Technology, Pingtung 912301, Taiwan; winton@mail.npust.edu.tw

³ General Research Service Center, National Pingtung University of Science and Technology, Pingtung 912301, Taiwan

* Correspondence: hwkuo@mail.npust.edu.tw; Tel.: +886-8-770-3202 (ext. 6224)

Abstract: The integration of smart sensor networks and Internet of Things (IoT) technologies has emerged as a key strategy for enhancing productivity and sustainability in agriculture and aquaculture under increasing climate and resource pressures. This review consolidates empirical findings on the performance of sensor-driven systems in optimizing the management of water, nutrients, and energy. Studies have demonstrated that IoT-based irrigation systems can reduce water use by up to 50% without compromising yields, while precision nutrient monitoring enables a 20–40% reduction in fertilizer inputs. In aquaculture, real-time monitoring and automated interventions have improved feed conversion ratios, reduced mortality by up to 40%, and increased yields by 15–50%. The integration of artificial intelligence (AI) into IoT frameworks further enhances predictive capabilities and operational responsiveness. Despite these benefits, widespread adoption remains constrained by high infrastructure costs, limited sensor robustness, and fragmented policy support. This paper provides a comprehensive evaluation of current technologies, adoption barriers, and strategic directions for advancing scalable, sustainable, and data-driven food production systems.

Keywords: climate resilience; technology adoption; digital transformation; precision monitoring; smart farming; artificial intelligence (AI); environmental stewardship; farm management; edge computing



Academic Editor: Jan Hopmans

Received: 25 April 2025

Revised: 1 June 2025

Accepted: 4 June 2025

Published: 6 June 2025

Citation: Liu, L.; Cheng, W.; Kuo, H.-W. A Narrative Review on Smart Sensors and IoT Solutions for Sustainable Agriculture and Aquaculture Practices. *Sustainability* **2025**, *17*, 5256. <https://doi.org/10.3390/su17125256>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Climate change has increasingly disrupted agricultural and aquacultural systems, directly threatening global food security. Extreme fluctuations in temperature have been shown to reduce crop yields by as much as 10%, while changes in rainfall patterns intensify water scarcity risks [1–4]. These climate-induced stressors also raise the frequency of disease outbreaks in terrestrial and aquatic environments, as evidenced by ecosystem instability and the emergence of harmful algal blooms that compromise aquaculture productivity [5–8]. Without effective adaptive strategies, projections suggest a marked rise in undernutrition rates driven by declining productivity in food systems [9,10]. This has led to a growing consensus on the necessity of implementing integrated, climate-resilient approaches to stabilize food production in vulnerable regions [11–14].

Traditional management practices in agriculture and aquaculture are increasingly inadequate under current environmental pressures. In agriculture, conventional irrigation

and fertilization practices accelerate soil degradation and water resource depletion, often resulting in nutrient runoff that contaminates aquatic systems [15,16]. Similarly, aquaculture practices lacking real-time monitoring tools frequently suffer from overfeeding, poor water quality control, and increased disease incidence [17,18]. Manual monitoring in both domains also leads to delayed decision-making and inefficient resource allocation, further reducing system efficiency [19,20]. In contrast, empirical studies have demonstrated that smart sensors and Internet of Things (IoT)-based systems can deliver timely data for automated interventions, reducing input waste while improving operational accuracy [21–24]. These technologies are thus positioned as viable alternatives to conventional practices, particularly in enhancing sustainability outcomes and production stability [25–27].

IoT and smart sensing technologies have emerged as transformative tools to address inefficiencies in agricultural and aquacultural production systems. When deployed across farms and aquaculture facilities, these systems enable real-time data acquisition and decision-making related to irrigation, fertilization, pest and disease management, and environmental control [28–32]. Research indicates that IoT applications enable the optimization of water usage, disease prevention, and more accurate yield forecasting by providing actionable insights derived from massive datasets collected through sensors strategically positioned across agricultural landscapes and aquaculture systems [33–36]. The integration of artificial intelligence (AI) further enhances these capabilities by identifying predictive patterns and facilitating automated interventions [36–38]. High-speed 5G networks and edge computing infrastructures enable low-latency communication between devices and localized processing of sensor data, ensuring system scalability and responsiveness [35,39]. Together, these technological frameworks offer the potential to transform conventional management practices into intelligent, data-driven systems that ensure productivity while safeguarding long-term environmental sustainability [40,41].

To contextualize these technological developments, this article presents a technology-oriented narrative overview of smart sensing and IoT applications in agriculture and aquaculture. Rather than conducting a systematic review with exhaustive statistical coverage, this study focused on synthesizing representative case studies and emerging implementation patterns that illustrate real-world applications and domain-specific challenges. The literature was retrieved from major scientific databases, including Web of Science, Scopus, and IEEE Xplore, using combinations of keywords such as “smart agriculture/aquaculture”, “IoT”, “precision agriculture/aquaculture”, “AI-based IoT (AIoT)”, and “sensor deployment”, linked with Boolean operators (e.g., AND and OR). The review scope was restricted to publications from 2014 to 2024 to ensure relevance to current trends and technologies.

This review aims to evaluate the practical applications and quantifiable benefits of smart sensors and IoT technologies in agriculture and aquaculture, focusing on resource efficiency, production gains, and environmental sustainability [42,43]. Second, it evaluates the technological, economic, and policy-related barriers that hinder large-scale adoption. These include sensor durability, interoperability challenges, high initial investment costs, and policy limitations. By identifying these barriers and examining documented case studies, this paper aims to provide strategic insights into the future development of smart, sustainable, and scalable IoT applications in food production systems [44,45].

2. Literature Review on Smart Sensors and IoT Applications in Agriculture and Aquaculture

2.1. Sensor Applications

2.1.1. Sensors Applications in Agriculture

- Environmental Monitoring Sensor

IoT-based environmental monitoring sensors in agriculture have significantly enhanced the capability to monitor and manage crucial environmental parameters, which is essential for optimizing agricultural practices. Notably, temperatures are measured using thermoresistive sensors, which detect resistance changes in a material as the temperature varies, while humidity is monitored through capacitive sensors that sense changes in dielectric properties. Both sensor types are key for early disease prediction and facilitating climate adaptation by allowing farmers to monitor microclimates closely, as specific temperature and humidity levels can trigger disease outbreaks [46,47]. Studies show that precise control of these environmental factors can result in improvements in crop yields, with some systems achieving up to a 25% increase over traditional methods due to their enhanced capacity for real-time data collection and response [48,49]. Similarly, solar radiation sensors, including photodiode and thermopile sensors, complement agricultural IoT systems by measuring solar radiation to optimize crop growth conditions and greenhouse management. Photodiode sensors detect light intensity through the photoelectric effect, producing signals in proportion to solar radiation. In contrast, thermopile sensors measure temperature gradients resulting from solar heating and convert these gradients into electrical signals. This distinction allows both sensor types to monitor the amount of solar energy plants receive, which is critical for maximizing photosynthesis and optimizing growing conditions by controlling light exposure [50,51]. Studies have shown that optimizing solar radiation exposure within greenhouses can significantly enhance crop yield, making these sensors essential tools in precision agriculture [52,53].

Another important aspect of IoT-based sensors in agriculture is the application of carbon dioxide (CO₂) sensors, particularly non-dispersive infrared (NDIR) sensors. These sensors operate by employing a light source and a gas detection chamber where CO₂ molecules absorb specific infrared wavelengths. The amount of light detected beyond the gas chamber correlates directly with the concentration of CO₂ in the environment. This measurement is critical in greenhouses, where optimizing CO₂ levels can enhance photosynthesis efficiency, reduce water use, and increase plant growth [54,55]. By enabling farmers to maintain optimal CO₂ concentrations, these sensors contribute to improved plant health and productivity while facilitating energy-efficient management of greenhouse environments [50,53]. By integrating these sensor technologies, IoT applications in agriculture enable automated climate control systems. By continuously monitoring temperature, humidity, solar radiation, and CO₂ levels, farmers can make informed decisions that drive effective disease risk assessments and optimize growing conditions throughout the crop cycle [48,49]. With the advent of IoT, farmers are equipped to engage in smart farming practices that dynamically respond to environmental changes, ensuring enhanced productivity amidst climate variability [56,57].

- **Soil Monitoring Sensor**

IoT-based soil monitoring sensors are pivotal in the realm of precision agriculture, as they enable real-time assessment of soil conditions, thus facilitating optimized irrigation and fertilization strategies. Among these, soil moisture sensors are of paramount importance. Three prevalent types are resistive, capacitive, and time-domain reflectometry (TDR) sensors. While resistive sensors infer moisture through changes in electrical resistance, they may be sensitive to soil salinity and variability [58,59]. TDR sensors offer more robust readings by measuring the time delay of electromagnetic pulses in soil, yielding accurate volumetric water content data. Research indicates that the application of TDR sensors can significantly improve irrigation scheduling, contributing to water usage reductions—by as much as 30%—while still achieving optimal crop yields [51,60]. Soil temperature sensors, mainly thermistors and thermocouples, play a critical role in managing crop root zone conditions. Thermistors provide precise temperature tracking, supporting the regulation

of microbial processes vital to nutrient cycling. Thermocouples, while less sensitive, are more durable under harsh field conditions [61,62]. Monitoring soil temperature not only influences irrigation practices but also helps optimize microbial activity that is essential for nutrient cycling, hence directly impacting crop productivity [63].

Regarding other soil indicators, electrochemical-based nutrient sensors have become increasingly significant. These sensors detect nitrogen (N), phosphorus (P), and potassium (K) concentrations through ion-selective electrodes (ISEs), which function based on the potentiometric measurement of ionic activity in the soil solution [60]. This technology allows for real-time monitoring of essential macro-nutrients, thereby informing precise fertilization practices that can enhance both crop yield and soil health [64]. Specifically, the integration of electrochemical sensors in an IoT framework facilitates automated precision fertilization, where nutrient application is regulated based on immediate soil nutrient status, minimizing over-fertilization and its associated environmental impacts [65,66]. By addressing both the physical and chemical properties of soil, these IoT-based sensors collectively enhance the precision and sustainability of modern agricultural practices.

- Crop Health Monitoring Sensor

The advent of IoT-based crop health monitoring sensors has significantly advanced the ability to manage crop conditions dynamically and efficiently. Among these sensors, leaf wetness sensors utilize changes in electrical resistance or capacitance to detect moisture levels on leaf surfaces, providing vital data that help in the early prevention of diseases. These sensors can signal when humidity levels are conducive to the proliferation of fungal pathogens, thereby allowing farmers to take timely actions such as adjusting irrigation schedules or applying fungicides to mitigate crop stress and prevent yield losses [67–69]. By leveraging these real-time data, agricultural practitioners can develop strategies that enhance plant health and minimize reliance on chemical treatments, thereby promoting more sustainable farming practices [70,71].

Another innovative class of sensors critical to crop health monitoring includes normalized difference vegetation index (NDVI), multispectral, and hyperspectral imaging sensors. By analyzing light reflectance patterns, these sensors help detect nutrient stress, monitor plant vitality, and identify early signs of disease. For instance, NDVI has been effectively employed for the early detection of nitrogen deficiencies in wheat crops, achieving accuracy levels exceeding 90% [69,72,73]. The ability to monitor these variations allows farmers to implement targeted fertilization strategies that optimize input costs while enhancing crop yields. Moreover, the integration of these imaging technologies into decision support systems enables precision agriculture to transition into an era of data-driven management, wherein farmers can make informed decisions based on empirical evidence rather than traditional heuristics [74,75]. Dendrometers, which utilize strain gauges or linear variable differential transformers (LVDT) technology, offer additional insights by measuring changes in the diameter of plant stems or trunks. These data can indicate growth rates and potential water stress, providing actionable intelligence that helps farmers manage irrigation more effectively [76,77]. Continuous growth monitoring enables timely responses to nutrient and water needs, transforming management from reactive adjustments to forward-looking strategies. Leveraging the capabilities of various types of sensors, from moisture detection to imaging and physical growth monitoring, provides a multifaceted approach to managing food production challenges [70,78,79]. By integrating real-time crop physiological indicators with environmental data, these sensor systems significantly improve yield prediction and harvest planning. Together with water management sensors, they enable a holistic decision support framework for optimizing productivity and resource efficiency [68,80,81].

- Water Sensor

In the realm of smart agriculture, the integration of IoT-based technologies has revolutionized water resources and irrigation management. Automated irrigation systems, driven by IoT-enabled actuators and flow sensors, utilize cutting-edge methodologies such as ultrasonic and electromagnetic flow meters. These systems precisely control irrigation activities by analyzing real-time soil water content data. By effectively calibrating water supply to match crop needs, IoT systems can achieve a significant reduction in water consumption, with reported efficiencies of up to 38% [82]. This data-driven approach fosters an environment where water resources are utilized judiciously, ensuring that crops receive adequate hydration while preventing over-irrigation, which can lead to waterlogging and plant health issues.

In tandem with automated irrigation, water level sensors also play a crucial role in optimal water distribution for agricultural irrigation systems. These sensors, based on pressure or ultrasonic technology, monitor water levels in reservoirs, contributing to balanced and judicious water management. The continuous feedback provided by these sensors allows for real-time adjustments to irrigation schedules, adapting to changing environmental conditions and further enhancing system efficacy for sustainable agriculture [3,4]. Such innovations are paramount, particularly in water-scarce regions where irrigation practices need to be meticulously managed to balance agricultural output with available resources.

Moreover, water quality monitoring sensors, utilizing electrochemical and optical fluorescence technologies, are instrumental in sustaining agricultural productivity. These sensors assess vital metrics such as pH levels, turbidity, and specific contaminants in water bodies accessible for irrigation. By providing consistent and real-time data regarding water quality, these sensors empower farmers to make informed decisions about water use, ensuring that agricultural practices do not compromise soil health or crop integrity [3,83]. Therefore, integrating water quality sensors within IoT frameworks supports the establishment of sustainable water management practices by ensuring that only water meeting quality standards is utilized in irrigation. By preventing over-irrigation and fostering precise water management, these advanced technologies facilitate improved crop productivity and resource allocation [84,85]. This not only results in cost-effective and resource-efficient farming but also serves as a crucial step towards the sustainable agricultural practices needed to secure food production for the future [86,87].

- **Pest Sensor**

IoT-based pest monitoring and smart agricultural machinery sensors are revolutionizing the efficiency and effectiveness of farming practices. Pest detection sensors, employing computer vision and infrared-assisted pheromone traps, have been integral in monitoring insect populations. Studies show that computer vision-based systems can achieve over 95% accuracy in pest identification, making them excellent for targeted interventions [88,89]. These systems analyze real-time field images to identify target pests, enabling precise interventions that reduce pesticide use and improve ecological outcomes [90].

On the other hand, infrared-assisted pheromone traps attract target pests using species-specific cues and monitor their presence through automated computer vision- or sensor-based detection [91]. This dual-technology approach presents a sustainable avenue for pest management, ultimately reducing reliance on broad-spectrum pesticides and promoting environmental health [92,93].

- **Smart Agricultural Machinery Sensor**

Automated fertilization sensors are equally pivotal in precision agriculture, utilizing technologies such as ISEs and electrochemical sensors to measure soil nutrient concentrations accurately. These sensors provide real-time data on nutrient levels, enabling farmers to adjust fertilizer applications based on crop needs rather than fixed schedules, often lead-

ing to over- or under-fertilization [94,95]. The dynamic adjustment of fertilizer application based on these real-time data optimizes nutrient delivery and significantly mitigates the risk of soil degradation and waterway pollution that can arise from excess fertilizer runoff [74]. Studies demonstrate that real-time monitoring leads to more efficient resource allocation, with some estimates showing a reduction of fertilizer application by up to 30% without compromising crop yield [94]. In conjunction with these advancements, smart agricultural machinery sensors, such as the global positioning system (GPS), light detection and ranging (LiDAR), and ultrasonic sensors, facilitate the automation of farming machinery, including tractors, drones, and robotic sprayers.

GPS technology enhances precision in field operations, allowing for accurate navigation and positioning of equipment, thereby enhancing productivity and minimizing overlapping applications of inputs, which leads to material savings [96]. LiDAR technology, utilized in conjunction with drones, provides high-resolution, three-dimensional mapping of crop fields, allowing farmers to assess crop health and detect pest infestations with remarkable accuracy [97]. Ultrasonic sensors enable robotic sprayers to assess the size and distance of objects, ensuring that pesticide applications are both targeted and efficient [98]. The integration of these technologies into autonomous systems allows for an unprecedented level of operational efficiency; studies indicate that these automated solutions can enhance labor productivity significantly [99,100]. This optimized fertilization regime, driven by IoT capabilities, aligns perfectly with sustainable agriculture to achieve higher productivity with a lower environmental impact.

2.1.2. Sensor Applications in Aquaculture

- Temperature Sensor

Temperature sensors are pivotal in aquaculture for providing accurate and real-time data on water temperatures, which directly influence fish growth, metabolism, and overall health. Common types include thermocouples and resistance temperature detectors (RTDs), which detect temperature changes through voltage or resistance variation [17,101]. These devices are crucial in both recirculating aquaculture systems (RASs) and open-water farms, where real-time monitoring is needed to maintain optimal operating conditions, ensuring that water temperatures are kept within a stable range of ± 1 °C [102]. Variations beyond the ± 1 °C threshold can lead to stresses that adversely affect growth and health, potentially resulting in decreased production and increased susceptibility to disease [103]. For instance, certain species of fish, such as the Atlantic salmon and Coho salmon, display significant variations in growth rates and feed conversion ratios when maintained at their optimal temperature ranges; studies indicate that growth can improve by up to 20% when the temperature is kept within this narrow band [104,105]. Moreover, research has highlighted that temperature fluctuations beyond this range can increase mortality, with temperature-related stress reported to cause a drop in survival rates by as much as 15% in juvenile fish under controlled aquaculture conditions [106,107]. Quantifying the impact of temperature control on aquaculture yields is imperative. Data suggest that for every 1 °C increase in temperature up to an optimal point, fish growth can improve by 0.5 g/day in species such as the Asian seabass [108]. Furthermore, a case study conducted within the context of RASs revealed that consistent temperature regulation had resulted in an increase in feed conversion ratios of up to 1.7, alongside a 30% reduction in feed wastage, demonstrating the tangible benefits of precise temperature control [109,110]. In open-water farms, this is corroborated by observed increases of up to 40% in biomass yields among species such as tilapia when strict temperature adherence is practiced [111].

In RASs, one effective implementation involved deploying an IoT-based monitoring system facilitating real-time adjustments to water temperatures and other parameters,

which ultimately enhanced fish survival rates by 25% [112]. Similarly, data gathered from open-water farms show that the application of advanced temperature monitoring systems has led to an increase in annual production yields by approximately 15%, showcasing the robust link between temperature management and aquaculture productivity [113]. The continuous assessment of water quality parameters, including temperature, through innovative sensor technologies has emerged as a cornerstone of effective aquaculture practices, benefiting both operational efficiency and fish well-being [114]. By ensuring stable temperature control, aquaculture operations can significantly enhance fish metabolic processes, immune responses, and overall feed efficiency, yielding quantifiable increases in harvest outputs.

- pH Sensor

pH sensors play a crucial role in aquaculture, leveraging the electrochemical properties of glass electrodes to monitor hydrogen ion activity in water. The fundamental operating principle of these sensors is based on the detection of voltage differences that correspond to pH levels. Regular calibration of these sensors with buffer solutions is essential to ensure accurate measurements. This calibration process helps mitigate errors caused by factors such as drift or temperature changes, thus promoting reliable monitoring, which is vital for maintaining optimal aquatic environments [35,115,116]. Maintaining optimal pH levels in aquaculture is imperative for the health of aquatic species. For most fish and crustacean species, the ideal pH range is between 6.5 and 8.5. Deviations from this range can induce physiological stress, metabolic disturbances, and increased susceptibility to diseases [117,118]. Studies have shown that maintaining pH stability within a narrow range of ± 0.3 can significantly reduce fish mortality rates by 15–25% while also improving growth rates by 10–18% [117].

Empirical evidence supports the effectiveness of IoT-based pH monitoring systems in aquaculture settings, particularly in shrimp and freshwater fish farming. These systems facilitate real-time monitoring of water quality, thereby preventing sudden fluctuations in pH that can lead to chemical imbalances and associated health issues in fish populations. For instance, a study highlighted the development of an automated IoT system that continuously assesses pH and temperature, allowing farmers to respond swiftly to adverse conditions [119]. Such technological advancements not only enhance the management of aquaculture environments but also contribute to higher yield and sustainability [116,120]. Thus, the integration of IoT-based technologies for monitoring pH levels exemplifies the ongoing shift towards data-driven aquaculture practices, underscoring a collaborative approach between traditional farming and modern technological capabilities.

- Dissolved Oxygen (DO) Sensor

DO sensors provide essential metrics for water quality management. Oxygen concentration in aquatic environments affects not only the health and growth of aquatic organisms but also their metabolism. Two major sensing technologies are commonly applied. Optical sensors, based on fluorescence quenching, measure changes in dye fluorescence caused by oxygen interaction, offering accurate and continuous data without oxygen consumption [121,122]. Electrochemical sensors, such as galvanic and polarographic types, rely on redox reactions at electrodes that produce a current proportional to the DO concentration, enabling cost-effective and field-deployable monitoring [123].

In the context of aquaculture, real-time monitoring of DO levels is fundamental in preventing hypoxia, a condition harmful to aquatic life that occurs when oxygen levels drop below critical thresholds, typically between 5–8 mg/L for most fish species [124]. This chronic condition can lead to reduced growth rates, increased stress, and heightened mortality rates among fish stocks. With continuous DO monitoring, aquaculture producers

can implement timely interventions, such as aeration or water circulation adjustments, to maintain optimal oxygen levels, thus optimizing the health and growth rates of fish populations [125]. Data from multiple studies indicate that the introduction of continuous DO sensors has substantially lowered hypoxia-induced mortality rates by as much as 30–50% while simultaneously improving feed conversion ratios (FCR) by 12–22% [126]. These quantitative benefits demonstrate how effective management of dissolved oxygen can enhance both the economic viability and sustainability of aquaculture practices. Moreover, empirical research indicates that fish farms employing these technologies have reported fewer disease outbreaks related to hypoxic conditions, further emphasizing the importance of dissolved oxygen monitoring for both animal welfare and commercial productivity [107], underscoring the essential role of these sensors in fostering sustainable fish farming practices worldwide.

- Ammonia and Nitrate Sensor

The effective management of nitrogen-based compounds, specifically ammonia (NH_3) and nitrate (NO_3^-), is crucial for sustaining aquatic life and ensuring optimal fish growth. These kinds of ISEs and spectrophotometric methods play an essential role in the detection of these nitrogen compounds. ISEs function based on the selective permeability and electrochemical properties of membranes, allowing for the direct measurement of NH_3 and NO_3^- concentrations in water samples by responding to the ionic activity of these analytes [127]. In contrast, spectrophotometric techniques involve a chemical reaction, which can then be quantitatively analyzed by measuring absorbance at certain wavelengths [127,128]. The choice of measurement method can significantly affect the accuracy and reliability of nitrogen detection in aquaculture settings.

Regular monitoring of nitrogenous compounds can help maintain these critical levels, thereby promoting healthy growth conditions and reducing stress on aquatic organisms. For instance, many fish species are at risk when NH_3 concentrations exceed 0.02 mg/L, which is considered the threshold for acute toxicity [129]. Studies have demonstrated that implementing real-time monitoring systems can effectively decrease toxic ammonia spikes by approximately 40–60%, leading to a concomitant reduction in disease outbreaks by as much as 20–35% [130,131]. Another case highlighted a fishery that experienced a reduction of 70% in nitrogen concentrations, corresponding to a notable improvement in fish health and growth rates, attributable to the timely adjustments made possible through real-time monitoring [132]. Similarly, real-time nitrate monitoring has been reported to delineate nitrate spikes effectively, leading to interventions that prevent spikes exceeding 5 mg/L, thus preserving the viability of aquatic life [128]. In a recent implementation in shrimp farms, the integration of a wireless sensor network facilitated the rapid detection of nitrogen levels, resulting in a decrease in ammonium concentrations by over 90% within days [133,134]. These measurements were crucial in maintaining operational health for sustainable shrimp farming [135]. Ammonia and nitrate sensors underscore a transformative progression in aquaculture, allowing for nuanced and real-time management of nitrogen levels. Their reliable operation forms the backbone of preventative measures against toxicity, subsequently fostering healthier aquatic environments and contributing to greater productivity within the industry.

- Salinity Sensor

Salinity sensors are critical in aquaculture for maintaining optimal water quality, especially in marine and brackish systems. Most rely on conductivity-based technology, where a voltage is applied across submerged electrodes to measure ionic current flow. This current reflects the water's electrical conductivity, a direct proxy for total dissolved solids and salinity levels. Typical sensors calibrated for sodium chloride solutions can

detect salinity across a broad range (1–100 parts per thousand, ppt), with a sensitivity slope of approximately 105 mV/dec [118]. This is particularly relevant for species such as *Litopenaeus vannamei*, for which optimal salinity levels range between 5 and 35 ppt, depending on specific growth conditions and developmental stages [136,137]. For example, it has been documented that ensuring optimal salinity levels can improve shrimp survival rates by 20–30% and enhance growth efficiency by 15–25% [138,139]. In shrimp hatcheries specifically, the application of robust salinity management with sensors has been linked to reduced mortality rates and more efficient feed conversion ratios, reflecting overall economic gains for aquaculture operators [140,141]. The abovementioned research has shown that when salinity was maintained at optimal levels conducive to these species, the survival rates were markedly improved, and the stock's overall health and resilience were enhanced as well [137,142]. Therefore, the integration of salinity sensors within aquaculture operations is essential for promoting both environmental sustainability and economic profitability.

- Oxidation-reduction Potential (ORP) Sensor

The measurement principle for ORP is based on the electrochemical behavior of platinum or gold electrodes. These electrodes measure the voltage difference in millivolts, allowing the assessment of the oxidation or reduction status of the water. This is essential for evaluating disinfection efficacy and the degradation of organic matter, notably in RAS and pond environments. Typically, an ORP reading above +350 mV indicates effective disinfection through agents such as ozone or chlorine, rendering the water relatively free of microbes, while lower values may suggest the presence of organic loading or pollutants detrimental to aquatic life [143].

Maintaining an ORP level within a specific range is crucial for enhancing the biofiltration process, which is vital for converting toxic ammonia into less harmful nitrate. In intensive farming systems, frequent ORP data collection has been associated with improved stock health and productivity, with some studies reporting growth rate increases of around 15–20% when optimal ORP levels are maintained [144]. A similar study on freshwater fish farms demonstrated that integrating ORP monitoring systems enabled responsive management of DO levels, significantly reducing mortality rates during warm weather; reductions were observed from above 15% to below 5% during peak heat periods [143]. In another case involving the deployment of ORP sensors in Pangasius farming in Vietnam, consistent monitoring showed a correlation between ORP levels and disease incidence. The analysis reported that regular ORP monitoring resulted in a 25% reduction in disease outbreaks attributed to improved water quality management, an essential statistic for supporting the rapidly expanding aquaculture sector in the region [145,146]. These empirical observations underline the necessity of integrated ORP monitoring systems in modern aquaculture, emphasizing their role in mitigating environmental risk factors and enhancing production efficiency while safeguarding aquatic health.

- Turbidity Sensor

Turbidity sensors play a pivotal role in aquaculture water quality management. Proper turbidity control is not merely a technical requirement; it is fundamental to maintaining the health and productivity of aquatic organisms. Typically, turbidity sensors measure water clarity using laser scattering and infrared detection. Suspended particles scatter the laser light, and the intensity of the scattered signal, captured by infrared sensors, reflects turbidity levels. High turbidity can hinder gill function, reduce oxygen uptake, impair feeding due to low visibility, and restrict light penetration needed for algal growth, affecting filter-feeding species [147,148]. Research has demonstrated that feeding rates decline significantly when

turbidity exceeds 25–70 NTU, while maintaining levels below 50 NTU can enhance shrimp productivity by up to 25% [149–151].

Field studies further underscore the practical value of turbidity monitoring. In RASs, for example, maintaining turbidity between 10–20 NTU has been shown to reduce stress markers and improve growth in species like pikeperch [150,152]. Similarly, in shrimp farming operations, consistent turbidity control has been linked to lower mortality rates, better growth performance, and improved resistance to disease. Importantly, real-time sensor feedback enables timely interventions, such as adjusting feed inputs or aeration, leading to up to a 40% reduction in feed waste [153,154]. This integration of turbidity sensors with IoT-based platforms is transforming aquaculture monitoring. By combining turbidity data with other environmental parameters, these systems facilitate predictive analytics and adaptive management strategies [17].

2.2. IoT Applications

The transformative impact of the IoT on modern precision agriculture is significant. By enabling real-time monitoring and wireless data transmission, the IoT facilitates the collection and analysis of agricultural data, allowing farmers to optimize their practices for enhanced yield and sustainability [84,155,156]. AI-driven analytics further elevate this capability, driving automated decision-making that optimizes input use, reduces waste, and ultimately supports a more sustainable agricultural framework [157,158]. Integrating various technologies, including sensors, drones, and data analytics, is pivotal to the precision agriculture paradigm, where efficiency and resource management are paramount [159–161]. These advancements in data-driven farming foster improved monitoring of environmental conditions and health, setting the stage for a detailed exploration of IoT operational mechanisms that enhance agricultural management practices [162–164].

2.2.1. Data Collection

The advent of IoT technology has significantly transformed agricultural practices, particularly through the deployment of sensor-based real-time monitoring systems. Central to these systems is an effective sensor network architecture comprising low-power embedded systems such as ESP32 and STM32 microcontrollers, which are adept at managing vast data streams from various environmental sensors. These microcontrollers enable the collection, processing, and transmission of data while ensuring energy efficiency, a critical aspect for devices operating in remote agricultural settings where power sources may be limited [165,166]. The signal processing techniques employed in sensor networks are equally vital for ensuring the accuracy and reliability of the environmental data collected. Techniques such as filtering, calibration, and noise reduction are employed to enhance data quality before it is transmitted for further processing.

Filtering is commonly used to eliminate outliers and reduce the effects of noise in sensor readings, which is essential given that environmental sensors are often exposed to various interferences [167,168]. Advanced calibration methods ensure that sensor outputs remain consistent with physical conditions, which is crucial for achieving high-quality measurements over time [169]. Organizational frameworks combining both fog and edge computing paradigms allow for collaborative data handling, where pre-processing at the edge leads to reduced data redundancy and more focused analyses on localized phenomena, thereby enhancing the efficiency of data processing routines across extensive networks of IoT devices [166,170,171]. Moreover, achieving precision in signal processing also involves the implementation of sophisticated algorithms capable of adaptive filtering and predictive modeling. These algorithms can dynamically adjust their parameters based on incoming data trends, ensuring that disruptions in data collection due to environmental factors are

accounted for in near real time [172–174]. The combination of these cutting-edge techniques not only enhances the reliability of environmental data collection but also yields actionable insights that are critical for optimizing agricultural and aquaculture practices in an era of increasing environmental variability and demand for sustainable solutions [165,175].

While the agricultural sector has experienced considerable advancements through IoT technologies, similar innovations have also catalyzed transformative progress in aquaculture, particularly in the domain of water quality monitoring and ecosystem management. In recent years, the integration of IoT technologies into aquaculture has significantly enhanced water quality monitoring, a critical factor for sustaining aquatic ecosystems and improving fish farming efficiency. Using an array of sensors, operators can continuously measure essential water quality parameters such as temperature, pH, turbidity, and dissolved oxygen, ensuring optimal conditions for aquatic life. A notable study demonstrated a real-time water quality monitoring system that could collect data every minute from multiple sensor nodes, aggregating up to 10,080 data points per sensor per week, which supports informed decision-making regarding water treatment interventions [18,176]. This is particularly vital in aquaculture, where minor fluctuations in water quality can result in significant losses, highlighting the critical role of real-time monitoring in preemptive management practices [120]. The implementation of these IoT-driven monitoring systems provides not only near-instantaneous access to data but also remote operability, allowing for effective water management without the immediate presence of personnel [177]. With the deployment of cost-effective sensors, many aquaculture facilities have reported an increase in both productivity and sustainability, citing operational cost reductions due to decreased human labor and improved resource management [178,179]. Data from such systems can be analyzed in real time, fostering an environment where timely corrective actions can be enacted to mitigate potential adverse impacts on fish health and growth [18].

2.2.2. IoT Communication

In the realm of precision agriculture and aquaculture, the effective transmission of data is crucial for optimizing operations. A comparative analysis of various wireless communication protocols reveals important distinctions in their operational characteristics. A long-range wide-area network (LoRaWAN) operates in the sub-GHz frequency bands (typically 915 MHz or 868 MHz) and boasts impressive power efficiency, consuming less than 100 mW while achieving transmission ranges of up to 10 km [180]. Similarly, narrowband IoT (NB-IoT) utilizes a licensed cellular spectrum, providing a data rate of approximately 55 kbps with moderate power consumption while supporting a range of around 10 km [181]. Conversely, 5G facilitates high-speed data transfer with latency as low as 1 ms, yet it demands higher power and is more suited for dense urban settings [180]. In comparison, Wi-Fi and Zigbee operate in the 2.4 GHz band, allowing for higher data rates (up to 600 Mbps for Wi-Fi) but limiting their transmission ranges to less than 100 m, making them less ideal for large agricultural fields [180,182]. On the other hand, while high-speed cellular networks such as 5G offer lower latency and a higher bandwidth, they face issues related to higher deployment costs and limited coverage in rural areas, impacting their scalability in broader applications such as precision agriculture [182].

Besides physical communication techniques, the data communication protocol plays a critical role in enhancing data transmission and interoperability across IoT ecosystems. Protocols such as message queuing telemetry transport (MQTT), the constrained application protocol (CoAP), and the hypertext transfer protocol (HTTP) contribute significantly to this framework. MQTT's lightweight nature is particularly advantageous for applications with limited bandwidth, making it suitable for regions with lower cellular connectivity [183]. Similarly, the CoAP is designed for constrained environments, allowing efficient communi-

cation between IoT devices and cloud platforms, thus enabling streamlined data processing and analysis in agricultural systems [184]. The HTTP continues to serve as a reliable protocol for tasks involving heavier data exchange, allowing integration with existing web services and applications for seamless user experiences [184]. The combination of these protocols underscores the potential for enhanced operational efficiency through effective data management in precision agriculture and aquaculture environments.

2.2.3. AI and Cloud Computing

The integration of AI and cloud computing in IoT-driven smart agriculture has made significant strides in recent years, primarily through the sophisticated architecture of cloud-based IoT platforms and advanced machine learning frameworks. These platforms, such as Amazon Web Services IoT (AWS IoT) and Google Cloud IoT, provide comprehensive environments for data ingestion, processing, and analytics.

AWS IoT facilitates the seamless connection of devices, ensuring efficient data flow from sensors in the field to cloud storage and processing units. The architecture typically incorporates multiple layers that manage device registration, connection handling, and data services, which collectively enable effective real-time data processing and analysis [185,186]. The data ingestion pipeline begins with the capture of sensor data, which is then processed through various services offered by these platforms, including data filtering and transformation tools. For example, AWS offers services such as AWS Lambda for serverless computing and AWS Kinesis for data stream processing, allowing for instant analytics that are crucial for timely decision-making in agriculture [187,188]. Google Cloud IoT similarly integrates AI capabilities within its platform, allowing users to apply trained AI models to real-time data streams [186]. The processing and analytics stages can involve sophisticated algorithms for time-series forecasting, anomaly detection, and reinforcement learning, implemented through popular frameworks such as TensorFlow (e.g., version 2.13.0) and PyTorch (e.g., version 2.0.1). In practice, predictive models deployed on cloud platforms often utilize long short-term memory (LSTM) networks for time-series forecasting of environmental variables and convolutional neural networks (CNNs) for image-based disease detection. These models require large volumes of labeled data for supervised learning, which may be sourced from UAV imagery, sensor logs, and farmer annotations. Cloud services such as AWS SageMaker or Google Vertex AI streamline this training process while enabling scalable deployment [189]. In the realm of predictive analytics, time-series forecasting methods can be fundamental for analyzing seasonal patterns in crop yields or predicting pest infestations. Anomaly detection plays a crucial role in identifying irregular patterns that could indicate problems in irrigation systems or soil health [190].

Additionally, the deployment of edge computing reduces latency, as data processing occurs closer to the source, enabling real-time analytics and decision-making capabilities that are vital for time-sensitive agricultural operations [191]. The integration of AI and cloud computing through these architectural frameworks and methodologies not only leads to enhanced predictive capabilities but also optimizes resource allocation and operational efficiency in smart agriculture. By analyzing vast amounts of data generated from sensor networks, cloud platforms facilitate informed decision-making, adaptive resource management, and strategic planning in agricultural practices, all while maintaining a keen focus on data privacy and security [192]. This technological alliance fosters an environment for sustainable agricultural practices, directly impacting food security and environmental management. Furthermore, the architectural design of these platforms allows for scalability, enabling systems to handle the growing number of IoT devices deployed across different agricultural sectors [193]. As the volume of data generated continues to rise, relying solely on cloud resources for data processing becomes impractical due to increased

latency and bandwidth constraints. By employing hybrid solutions that combine cloud and edge resources, agricultural stakeholders can achieve a more effective balance between computational demands and data privacy, thus ensuring the longevity and adaptability of smart agriculture systems [191].

In addition to smart agriculture, significant progress has also been made in aquaculture, particularly in the areas of environmental monitoring, predictive analytics, and operational automation. The advent of IoT technologies has catalyzed significant advancements in aquaculture, particularly through the integration of AI-based applications that enable enhanced monitoring and management of aquatic environments. These advancements are evidenced by the continuous collection of substantial datasets that allow aquaculture practitioners to leverage insights derived from machine learning algorithms, which can predict water quality and optimize fish health and feed usage. AI-powered sensory networks have proven effective in monitoring environmental conditions. These systems also support predictive analytics, allowing aquaculture operators to intervene before disease outbreaks escalate [23,178,194–196]. Additionally, emerging techniques in computer vision show promise in developing automated solutions for tasks such as fry sorting and feed waste identification, providing significant labor savings and efficiency improvements in aquaculture operations [121]. The fusion of big data and AI further catalyzes innovation within the sector, propelling aquaculture practices towards what is commonly termed “Aquaculture 4.0”. Incorporating advanced AI models has the potential to harness data from diverse sources, contributing to a holistic understanding of the aquatic environment [197] and positioning them favorably in a competitive market characterized by fluctuating demands for seafood products [24].

2.2.4. Automated Actuation

Automated actuation is a core technological advancement within IoT-based systems, enabling real-time responsiveness through the integration of sensing, decision-making, and mechanical execution. The architecture of these systems generally includes three key components: (1) sensor modules that monitor environmental parameters such as soil moisture, temperature, humidity, pH, and dissolved oxygen; (2) microcontroller units (MCUs) such as Arduino, ESP32, or STM32, which process data locally and serve as the system’s central logic unit; and (3) actuators, such as solenoid valves, servo motors, peristaltic pumps, or relays, which physically implement the desired actions. These components communicate via IoT protocols (e.g., LoRa, ZigBee, and MQTT over Wi-Fi or cellular) and are often supported by edge computing devices that execute control algorithms near the data source to reduce latency. The architecture may also integrate cloud-based services for remote monitoring, historical data analysis, and large-scale farm management [198–200].

The decision-making process within automated actuation relies heavily on control algorithms. Traditional control techniques, such as proportional–integral–derivative (PID) controllers, are widely used for maintaining system stability, especially in climate control applications. However, due to the complex and non-linear nature of agricultural and aquacultural environments, more sophisticated approaches have been adopted. Fuzzy logic controllers (FLCs) are capable of modeling imprecision and human-like reasoning, making them suitable for systems requiring multi-variable inputs and heuristic rules. Reinforcement learning (RL) and deep reinforcement learning (DRL) offer adaptive optimization by continuously learning optimal control strategies through environmental interactions. These AI-based algorithms can adjust irrigation or feeding schedules dynamically based on weather patterns, crop growth stages, or fish behavior patterns [7,198,201].

In agriculture, automated actuation systems are frequently deployed in precision irrigation and fertigation systems, where soil moisture sensors trigger real-time valve

control to deliver water or nutrients only when and where needed. For instance, fuzzy logic systems have been employed to regulate greenhouse temperature and humidity based on sensor thresholds, while machine learning models predict evapotranspiration rates to schedule irrigation more accurately. Multi-layered system designs typically include sensor nodes, a local control unit, wireless transmission modules, and a cloud dashboard, allowing farmers to monitor and intervene remotely. These systems have demonstrated reductions in water and fertilizer use while also improving labor efficiency and promoting healthier crop development, particularly under variable climatic conditions [202–204].

In aquaculture, actuation systems enable automated feeding, water quality regulation, and waste management. Smart feeders, guided by sensors monitoring dissolved oxygen, temperature, and turbidity, initiate feeding events aligned with fish activity levels to improve feed conversion ratios and minimize waste. RASs use actuators to control aerators, pumps, and filters, often governed by fuzzy logic or machine learning-based controllers to maintain optimal conditions. These systems can reduce feed wastage by up to 30%, cut labor costs by 25%, and lower nutrient discharge by over 70%, supporting both productivity and environmental sustainability [41,205,206]. As automated actuation becomes increasingly embedded in IoT frameworks, its role in advancing sustainable agricultural and aquacultural practices is expected to expand significantly.

3. Summary of the Benefits of Applying IoT-Based Systems in Agriculture and Aquaculture

This section highlights the practical benefits of IoT-based systems from the perspective of agricultural and aquacultural decision-making. In response to the increasing complexity of resource management, the deployment of real-time sensing, AI-driven analysis, and automated response mechanisms has provided farmers and fish farmers with timely, data-informed tools that support more efficient, sustainable, and adaptive operations. The following two subsections focus on agriculture and aquaculture, respectively, providing empirically supported examples of how these technologies translate into concrete decision-making benefits, such as optimizing irrigation, fertilizer use, feeding schedules, and disease management.

3.1. Benefits of Smart Agriculture

The application of IoT-enabled precision agriculture has led to substantial improvements in several key agricultural metrics, driven by empirical data from various studies indexed in reputable sources. For instance, in terms of water conservation, IoT-based automated irrigation systems can reduce water consumption by 30–50% while simultaneously maintaining or enhancing crop yields by 15–20% [207,208]. A case study involving an automated irrigation system installed in Texas reported a 50% reduction in water usage, coupled with a 20% increase in yields for cotton crops, evidencing the dual benefit of conserving resources while boosting production [208]. This case underscores smart irrigation's capacity to balance water efficiency with yield optimization in arid regions. The efficient use of fertilizers is another significant advantage offered by IoT-enabled precision agriculture technologies. By employing precision nutrient management through IoT sensors, agricultural practices have seen a decrease in fertilizer waste by 20–40%, resulting in a 10–18% increase in nitrogen use efficiency [209,210]. For example, a study in California found that the integration of IoT sensors in nutrient application reduced fertilizer expenditures by approximately 25% while achieving yields similar to or better than traditional methods [210]. The data suggest that precision agriculture not only mitigates costs related to over-fertilization but also reduces the environmental impact of excess nutrient runoff.

Furthermore, the enhancement of crop yields through the implementation of smart technologies is noteworthy. AI-driven disease detection and sensor-based pest monitoring have led to increases in yields ranging from 12–30% across various crops [44]. In a specific case involving a greenhouse for tomato cultivation, the deployment of advanced pest management systems reduced pest damage by 30% and subsequently increased yields by approximately 18% [211]. Such advancements underscore the potential for precision agriculture to boost food production, especially amidst significant rising global demand. The capability for early disease detection is also a crucial component of precision agriculture. Smart sensing technologies have been reported to detect plant diseases 5–10 days earlier than traditional methods, reducing crop loss by 15–35% [209,212]. For instance, a vineyard that integrated an IoT-based health monitoring system recorded a 25% decline in crop loss due to early intervention, thus demonstrating the effectiveness of technology in safeguarding harvests [212]. This early warning capability exemplifies how digital interventions can transform agricultural practices by enabling proactive disease management. In summary, the quantified benefits of IoT-enabled precision agriculture, supported by empirical studies, reveal significant improvements in water conservation, fertilizer efficiency, crop yield enhancement, and disease detection. These synergistic advantages not only bolster agricultural productivity but also align with sustainable farming practices that are essential in an era marked by climate change and resource scarcity.

3.2. Benefits of Smart Aquaculture

The integration of IoT technologies in aquaculture has demonstrated measurable improvements in operational efficiency, mainly through the use of real-time sensing, intelligent feeding systems, and adaptive management protocols. Empirical studies have shown that continuous monitoring of temperature, dissolved oxygen, and turbidity enables dynamic adjustment of feed delivery, reducing waste and improving feed conversion ratios. Yield increases ranging from 15% to 50% and reductions in feed loss by 20–30% have been reported, with improvements largely attributed to machine learning-based decision support systems that respond promptly to environmental fluctuations [120,213]. In parallel, health monitoring via biosensors allows early detection of physiological stress, enabling timely intervention before disease outbreaks escalate. Studies have indicated that such systems can reduce mortality by approximately 40%, significantly lowering antibiotic use and contributing to improved food safety and animal welfare [196,214,215].

In addition to optimizing feed efficiency and fish health, IoT systems have also shown measurable impacts on environmental outcomes, particularly through improved water reuse and discharge management. Smart control systems equipped with ammonia, nitrate, and ORP sensors have been used to optimize water exchange, aeration, and filtration, achieving water reuse rates exceeding 90% and mitigating nutrient discharge into surrounding ecosystems [216]. These improvements also facilitate nutrient recycling and sludge reduction, promoting more sustainable closed-loop systems. In the face of climate variability, IoT-enabled environmental sensing supports adaptive capacity by detecting anomalies such as salinity shifts or hypoxic events. For instance, real-time temperature monitoring has been associated with a 25% increase in fish survival rates under heatwave conditions, reflecting the potential of such systems to mitigate risks driven by climate change [217–219].

Economic assessments have further supported the value of IoT technologies in aquaculture. Facilities integrating real-time analytics and automation have reported cost reductions of up to 50%, driven by improved feed efficiency, lower energy use, and reduced labor demands [220]. While these outcomes contribute to short-term efficiency, their long-term viability remains contingent on sustained investment, particularly in infrastructure and

training—areas where many aquaculture enterprises still face significant constraints. These findings underscore the IoT’s transformative potential in fostering sustainable intensification, enabling aquaculture systems to respond more effectively to economic pressures and ecological challenges.

3.3. Integrated Impact Assessment of the IoT in Agriculture and Aquaculture

A comparative analysis offers further insight into how these technologies create strategic value under varying environmental, economic, and policy conditions. To structure this synthesis, three widely used analytical frameworks are applied: PEST analysis (political, economic, social, and technological), 4P analysis (product, price, place, and promotion), and the value proposition canvas, which collectively provide a multidimensional view of technological integration across domains. These methods are employed not as theoretical subjects in themselves but as organizational scaffolds to map insights already supported by the literature cited in previous sections.

From a macro-level PEST perspective (Table 1), both sectors benefit from supportive policy trends and increased societal awareness of digital agriculture and food security. However, technological and economic factors remain uneven. In agriculture, market-ready solutions such as automated irrigation and nutrient management systems have demonstrated quantifiable resource savings, reducing water usage by 30–50% and fertilizer waste by 20–40% [207–209]. In aquaculture, real-time environmental monitoring has enabled feed optimization and reduced mortality rates by up to 40% [112,123,196]. Nonetheless, both sectors face challenges related to sensor durability, connectivity in remote environments, and the high initial investment required for infrastructure, especially among smallholder operations [58,59,123].

Table 1. PEST analysis of IoT applications in agriculture and aquaculture.

Aspect	Agriculture	Aquaculture
Political	Precision irrigation and disease forecasting align with national goals for water-saving and low-carbon agriculture [207,209].	Automated water quality monitoring addresses wastewater discharge and antibiotic regulation policies [214,215].
Economic	Water use is reduced by 30–50%, yield is increased by 15–35%; fertilizer waste is cut by 20–40%, and nitrogen use efficiency is improved by 10–18% [208–210].	Feed waste is reduced by 30%, productivity is improved by 20–50%, and mortality is reduced by 20–40% [112,123,196,214,221].
Social	Lower chemical input and nutrient runoff, supporting food safety and farmer adoption [209,212].	Reduced antibiotic dependence, improving food quality and consumer trust [215,216].
Technological	AI disease detection occurs 5–10 days earlier; leaf wetness sensors reduce crop losses by 15–35% [62,212].	Early-warning systems reduce disease-related mortality by 20–40%; integration challenges include sensor failure and model uncertainty [62,123,196].

From a 4P Analysis perspective (Table 2), IoT solutions in both sectors prioritize performance-driven outcomes, such as increased yields (up to 35% in agriculture and 50% in aquaculture) [44,112,213,221], but vary significantly in terms of accessibility and user adaptability. While agricultural technologies increasingly incorporate AI-driven diagnostics and user-friendly interfaces, aquaculture tools often require more technical expertise for multi-parameter calibration and integration with biological feedback loops. This gap underscores the importance of simplifying deployment pathways and investing in interface-level improvements that accommodate varying literacy and training levels.

Table 2. 4P analysis of IoT applications in agriculture and aquaculture.

Aspect	Agriculture	Aquaculture
Product	IoT-based irrigation, nutrient control, and AI pest detection [209–211].	Water quality sensors (DO, pH, and NH ₃) and smart feeding systems [112,214].
Price	Initial cost high, offset by resource savings and yield gains [208].	High investment, justified by waste reduction and improved yields [123].
Place	Applied in open fields, greenhouses, and high-tech farms [210].	Used in ponds, recirculating aquaculture systems, and hatcheries [196].
Promotion	Government subsidies and pilot farms drive adoption [51,208].	Public–private partnerships and regional demonstrations encourage use [41,215].

When assessed using the value proposition canvas (Table 3), IoT technologies offer clear functional and economic benefits, yet emotional and social dimensions are less consistently addressed. For producers, early disease detection and environmental alerts translate into tangible security, yet the lack of community-based deployment models or participatory design often limits widespread trust and local adaptation. Furthermore, while empirical evidence supports productivity and sustainability gains, the translation of sensor data into actionable insights remains a significant barrier to maximizing perceived value at the farm or pond level [23,210].

Table 3. Value proposition analysis canvas of IoT applications in agriculture and aquaculture.

Aspect	Agriculture	Aquaculture
Benefit	30–50% water savings, 12–30% yield improvement, and reduced pest/disease losses by 15–35% [44,207,211,212].	20–50% higher productivity, reduced feed waste, and improved survival rates [112,214,221].
Differentiation	Combines environmental sensing and AI for early alerts; 25% lower fertilizer cost and higher nutrient efficiency [209,210].	Integrates multiple sensors for real-time water monitoring and disease prevention [123,196].
Intangible Value	Enables ESG-compliant sustainable farming; reduces environmental degradation [51,207].	Reduces antibiotic use and pollution risk; improves environmental and public health perception [41,215,216].

The findings indicate that integrating the IoT into agriculture and aquaculture offers substantial benefits in resource efficiency, biological risk management, and productivity stabilization. Precision irrigation and nutrient systems have shown reductions of 30–50% in input use and yield increases of 15–35% in agricultural settings, while aquaculture systems leveraging real-time monitoring and smart feeders achieve reductions of up to 30% in feed waste and production gains of up to 50% [51,58,213]. Additionally, early detection enabled by sensing technologies has reduced losses due to disease and environmental stress by 15–40% across both domains [62,123,212,214].

These patterns reveal a structural convergence between the two sectors, driven by shared reliance on real-time feedback mechanisms, automation, and predictive analytics. Nonetheless, real-world deployment remains constrained by challenges such as sensor drift, interpretive complexity, and high initial investment, particularly for smallholders [58,59,123,210,222]. Moreover, the rise of AI-integrated IoT systems promises economic and operational improvements, yet also introduces new challenges, including algorithmic bias, hardware limitations, and integration complexity [70,123,191,222]. These issues highlight the importance of broader infrastructural, policy, and institutional support. The synthesis presented here establishes a foundation for subsequent examination of sensor per-

formance, economic feasibility, AI deployment constraints, and enabling policy conditions across both sectors.

4. Systematic Evaluation and Discussion

4.1. Technological Benefits and Challenges

4.1.1. Comparative Benefits of the IoT in Agriculture and Aquaculture

The adoption of IoT technologies in agriculture and aquaculture has led to substantial improvements in resource efficiency, the early detection of biological stressors, and overall productivity enhancement. In agriculture, empirical studies have shown that the use of capacitive and TDR sensors for precision irrigation can reduce water usage by 30–50% while simultaneously increasing crop yields by 15–35% compared to conventional methods [51,58,207]. Likewise, in aquaculture, IoT-driven monitoring systems that track dissolved oxygen, ammonia, and temperature have enabled dynamic feed adjustments and environmental control, resulting in reductions in feed waste of up to 30% and productivity gains between 20% and 50% [213,221]. Despite these benefits, several technical limitations persist, particularly in calibration drift, reduced reliability under harsh environmental conditions, and the high initial cost of infrastructure deployment, which collectively hinder adoption, particularly for small-scale producers [58,59,123].

In terms of biological risk mitigation, IoT systems have demonstrated notable advantages in both sectors. Agricultural applications employing leaf wetness sensors and hyperspectral imaging technologies have enabled early disease detection several days before symptoms are visible, reducing yield losses by 15–35% through timely responses [62,212]. Parallel developments in aquaculture indicate that continuous monitoring of water quality parameters, such as dissolved oxygen, ammonia, and pH, supports early intervention against disease outbreaks, decreasing mortality rates by 20–40% [123,196,214]. Nevertheless, real-world implementation remains challenged by predictive model uncertainty under rapidly fluctuating conditions and sporadic sensor failures, which interrupt data continuity and weaken system reliability [62,196].

While these technical hurdles remain, field-level observations continue to reveal productivity gains in both domains, reinforcing the case for IoT adoption. In agriculture, automated nutrient management systems have reduced fertilizer wastage by 20–40% and improved nitrogen use efficiency, contributing to crop yield increases of up to 18% [209,210]. In aquaculture, similar improvements stem from enhanced feed conversion and environmental control, which support yield gains exceeding 20% [23,112]. However, integrating multi-sensor data into actionable insights remains a practical barrier. Users frequently encounter challenges interpreting complex analytics and accommodating environmental variability, which impairs the effectiveness of data-driven management at the operational level [210]. Adding to these challenges are the energy demands of continuous sensing and wireless transmission, especially in remote or infrastructure-poor areas, representing an additional operational cost and sustainability concern.

Lastly, environmental sustainability represents a shared benefit across sectors, with IoT systems helping to minimize water consumption, nutrient leaching, and pollution. In agriculture, the use of precision irrigation and fertilization directly reduces ecological degradation linked to traditional practices [51,207]. In aquaculture, sensor-based control of nutrient discharge and reduced dependence on antibiotics address both environmental and public health risks [41,215,216]. Despite these advances, sustained environmental gains are contingent upon several enabling factors, including farmer competence, infrastructure support, and coherent policy frameworks. Without adequate training and socio-economic incentives, the broader sustainability potential of the IoT may remain unrealized [41,207]. Despite the sector-specific differences and implementation barriers outlined above, the

strategic landscape of the IoT in agriculture and aquaculture can be more holistically understood by synthesizing its internal and external factors. To that end, a SWOT analysis was conducted to systematically summarize the strengths, weaknesses, opportunities, and threats associated with IoT adoption across both sectors (Table 4). This comparative framework serves to reinforce the empirical findings while clarifying the practical and strategic dimensions that influence real-world deployment.

Table 4. SWOT analysis of IoT applications in agriculture and aquaculture.

Category	Agriculture—IoT	Aquaculture—IoT
Strengths [51,58,62,112,123,207,209,213,214]	- Improved water and fertilizer efficiency (up to 50%) - Early disease detection via leaf wetness and imaging sensors - Enhanced productivity and reduced nutrient leaching	- Real-time water quality monitoring - Optimized feeding (up to 30% waste reduction) - Reduced mortality through early alerts
Weaknesses [58,59,123,210,222]	- Sensor accuracy affected by soil salinity, temp, and drift - Complex analytics interpretation at field level - High initial costs for smallholders	- Biofouling and drift in DO/pH sensors - Frequent recalibration needs - Sensor failures disrupt control loops
Opportunities [70,71,209,223,224]	- AI-enhanced decision systems (agri-AIoT) - Policy incentives for precision agriculture - IoT-as-a-service models	- Aqua-AIoT for dynamic feeding and stress prediction - Regulatory compliance for water quality - Integration with blockchain or remote sensing
Threats [60,191,222,225]	- Connectivity limitations in rural areas - Lack of IoT standardization - Data privacy and governance concerns	- Similar infrastructure and standardization barriers - High cost of robust sensors - Uneven digital capacity among producers

On the strength side, IoT implementation has clearly advanced input efficiency, productivity, and early risk detection. Empirical evidence confirms that smart irrigation and aquaculture sensing systems can reduce water and feed consumption by 30–50% and simultaneously boost yields by up to 50% [51,58,213]. These improvements, alongside environmental benefits such as reduced pollution and nutrient leaching, reflect a shared potential for more sustainable production [41,207,215].

However, technical weaknesses such as calibration drift, environmental sensitivity, and user inexperience with multi-sensor systems continue to affect sensor reliability and data interpretation [58,123,210]. Moreover, high initial investment costs and limited digital literacy among smallholders further constrain adoption, particularly in developing contexts [59,222].

Opportunities are emerging with the integration of AI, precision analytics, and policy-driven incentives. The rise of agricultural AIoT (agri-AIoT) and aquacultural AIoT (aqua-AIoT) systems promises improved economic viability and broader accessibility via automated prediction and decision support platforms [70,71,209,223]. Financial models such as IoT-as-a-service also provide new avenues to lower entry barriers [125,224].

Conversely, external threats such as infrastructure limitations, data governance concerns, and lack of standardization across platforms remain critical barriers to scalability and interoperability [60,191,222]. These challenges underscore the need for not only technical

innovation but also coordinated policy and institutional support to realize the full potential of IoT-enabled systems.

4.1.2. Sensor Performance and Data Accuracy

Sensor calibration drift and interference from environmental variables represent primary accuracy challenges in agricultural IoT applications. For instance, Prasad et al. reported that resistive soil moisture sensors frequently experience measurement inaccuracies due to soil salinity and temperature fluctuations, with errors sometimes exceeding $\pm 10\%$ of actual moisture levels. Such inaccuracies can lead to inefficient irrigation strategies, resulting in over- or under-irrigation and subsequent reductions in crop productivity [58]. Similarly, sensor studies conducted by Teixeira and Santos emphasized that capacitive and TDR sensors, although generally more accurate than resistive sensors, can still encounter performance degradation under highly saline or clay-rich soil conditions, necessitating careful and regular recalibration [59]. These agricultural sensor limitations critically underscore the need for rigorous site-specific calibration protocols to mitigate potential adverse impacts on crop management decisions.

In aquaculture, sensor performance issues similarly constrain IoT efficacy, particularly due to biofouling and sensor drift. Xia et al. noted that electrochemical DO sensors typically exhibit weekly deviations of approximately 0.05–0.15 mg/L due to biofilm accumulation, requiring frequent recalibration and maintenance to maintain measurement reliability [123]. Such inaccuracies have direct implications, including delayed interventions, suboptimal oxygen management, increased fish stress, and substantial economic losses due to elevated mortality rates or growth impairments [196,214]. Electrochemical-based nutrient sensors, although crucial for real-time water quality management, can also suffer from reduced accuracy over time due to electrode deterioration and environmental interferences, requiring regular calibration to ensure accurate nutrient monitoring in intensive aquaculture systems [112,213].

To address these sensor accuracy issues, the prior literature consistently recommends adopting advanced sensor technologies tailored to challenging environmental conditions. Within agricultural contexts, capacitive and TDR-based soil moisture sensors demonstrate enhanced robustness and lower susceptibility to environmental interferences compared to traditional resistive sensors, albeit at higher initial investment costs [51,58]. Similarly, aquaculture's optical dissolved oxygen sensors and ISE-based pH sensors have proven more resistant to biofouling and drift than electrochemical sensors, maintaining high data accuracy with less frequent maintenance demands despite their increased initial expense [23,123]. Nevertheless, previous studies critically indicate that the high cost associated with these advanced sensor solutions can pose substantial barriers for smaller-scale operations, thus limiting widespread adoption without appropriate economic or policy incentives [41,210].

A crucial consideration frequently emphasized in existing research is the role of regular calibration protocols, maintenance schedules, and data validation techniques in ensuring long-term sensor reliability. Agricultural studies highlight that automated calibration systems and rigorous sensor validation methods, such as adaptive filtering and machine learning-based error detection algorithms, significantly enhance data accuracy and reduce decision-making uncertainties [62,226,227]. Likewise, the aquaculture literature strongly supports implementing automated sensor-cleaning devices and predictive maintenance algorithms to mitigate biofouling and drift-related inaccuracies [196].

4.1.3. Economic Feasibility

A growing body of economic evidence suggests that precision agriculture powered by IoT yields tangible cost savings while enhancing productivity. For instance, IoT-enabled irrigation systems consistently reduce water consumption by up to 50%, simultaneously lowering operational costs and enhancing profitability compared to traditional methods [36,51,58]. Similarly, precise nutrient management facilitated by IoT sensor networks can decrease fertilizer expenditures by approximately 20–40%, offering immediate operational savings while improving overall crop yields [60,228,229]. Comparable economic advantages are evident in aquaculture. IoT applications in water quality management, particularly those monitoring DO, pH, ammonia, and nitrates, have markedly improved feed conversion rates and growth performance, significantly reducing feed wastage and disease-related costs [120,123,221]. Automated feeding systems supported by the IoT further enhance economic viability, reducing feed expenses by approximately 15–30% while simultaneously increasing production outputs and profitability [205,213,223]. These outcomes collectively demonstrate that IoT adoption in both agriculture and aquaculture can yield strong returns, typically positive economic returns within a few production cycles [120,221].

However, despite these economic benefits, initial costs remain significant barriers, particularly for smaller-scale operators. High initial investments, including the procurement of precision sensors, the establishment of communication networks, and analytics infrastructure, can often exceed financial capabilities, especially in resource-constrained regions [222,224]. Additionally, ongoing maintenance costs stemming from sensor calibration drift, biofouling, and replacements significantly increase annual operational expenses by approximately 10–20%, further diminishing economic attractiveness [123,221]. To address these economic limitations, the integration of AI with IoT—referred to as agri-AIoT and aqua-AIoT—offers substantial potential. Agri-AIoT systems using AI-driven analytics for disease forecasting and yield prediction have demonstrated significant reductions in crop losses and input costs, leading to tangible economic benefits [70,71,209,212]. Similarly, aqua-AIoT approaches employing AI for automated feeding and proactive environmental monitoring consistently report enhanced productivity, improved feed efficiency, and overall economic returns [162,205,223]. Despite these promising outcomes, practical implementation still demands continuous technical support, including software maintenance and updates by skilled engineers, which may impose additional costs and limit adoption among small-scale producers.

Nevertheless, AI integration into IoT also introduces additional economic complexities, such as increased initial expenses for computing infrastructure, software, and specialized training, potentially elevating upfront costs by approximately 10–20% compared to standard IoT deployments [191,222]. Despite these higher initial investments, economic analyses consistently indicate positive medium-to-long-term returns due to improved decision-making precision, minimized resource waste, and reduced operational uncertainty [162,209,212]. Therefore, strategic actions, including innovative financing models (e.g., IoT-as-a-service), targeted governmental subsidies, and investments in rural infrastructure, are crucial to overcoming economic barriers and promoting widespread adoption [125,222,224]. Although IoT and AI-enhanced IoT (agri-AIoT/aqua-AIoT) technologies exhibit substantial economic advantages in agriculture and aquaculture, comprehensive strategies addressing initial and recurrent costs, infrastructure improvements, and policy support remain essential for realizing their full economic feasibility.

4.2. Future Developments and Strategic Applications

4.2.1. AI and Machine Learning in IoT-Based Decision-Making

Building upon the economic and operational benefits of agri-AIoT and aqua-AIoT introduced earlier, this section further discusses the technological advancements, challenges, and strategic implications associated with integrating AI into IoT-based decision-making systems. AI has increasingly become essential for improving operational efficiency, predictive accuracy, and resource use precision across both the agriculture and aquaculture sectors. In agriculture, AI-enhanced image recognition through advanced deep learning algorithms, such as CNNs, demonstrate notable accuracy rates surpassing 90% in disease detection, substantially outperforming conventional visual assessment methods [70,71,209,212]. Such high-accuracy systems facilitate early, targeted interventions, effectively reducing crop losses by approximately 30–35% and significantly improving yield through precise agrochemical applications [31,61,208,230]. Similarly, AI-driven automated feeding systems in aquaculture use IoT-collected behavioral and environmental data to optimize feeding schedules and quantities. Empirical studies validate substantial yield increases near 30%, accompanied by reduced feed wastage and enhanced economic viability [162,205,223].

Despite the demonstrated economic and operational advantages of agri-AIoT and aqua-AIoT, significant technological barriers remain, particularly regarding effective AI integration into existing IoT frameworks. Computational constraints represent a critical challenge, as real-time AI analytics typically depend on cloud infrastructures that often suffer from latency and unstable connectivity in rural agricultural and remote aquacultural contexts [191,224]. Edge computing has emerged as a promising alternative, enabling local processing and timely decision-making, yet its deployment remains restricted by limited hardware capabilities, power constraints, and algorithmic complexity. Thus, future research must prioritize developing lightweight ML algorithms explicitly tailored to edge environments, ensuring reliability under resource-limited conditions.

Algorithmic bias and data privacy concerns further complicate AIoT adoption. AI systems require extensive, representative datasets for effective training; however, existing datasets often reflect regional or crop-specific biases, limiting their broader applicability and accuracy across diverse environmental conditions [131]. Moreover, privacy apprehensions surrounding sensitive agricultural and aquacultural data significantly deter producers from adopting IoT systems, necessitating the integration of privacy-preserving technologies such as federated learning to build user trust and enable secure data utilization. Although federated learning offers decentralized data processing to address these concerns, substantial challenges persist, particularly regarding data heterogeneity and algorithm validation [61,62]. Additionally, sensor hardware and interoperability challenges significantly constrain broader AIoT integration. Agricultural IoT sensors frequently face severe environmental stresses, including soil moisture variability, extreme temperatures, and mechanical disturbances [51,58]. In aquaculture, IoT sensors must consistently operate under harsh aquatic conditions, encountering biofouling, corrosion, and prolonged calibration drift [123,221]. Moreover, the lack of universally recognized IoT communication standards increases technological fragmentation, complicating integration and scalability [60,222]. Therefore, addressing hardware durability requirements and standardizing communication protocols remain critical priorities for achieving broader and economically feasible IoT adoption.

A comparative analysis of agriculture and aquaculture reveals both shared objectives and operational differences. Both sectors prioritize resource optimization and predictive analytics, yet agricultural IoT systems must accommodate considerable spatial and temporal variability, such as diverse soil types and seasonal climate fluctuations [51,231]. Conversely, aquaculture IoT deployments emphasize continuous, high-frequency monitoring

and rapid response actions to sustain stable aquatic environments [112,221]. Recognizing these distinct operational contexts underscores the importance of tailored AIoT development strategies, emphasizing adaptable agricultural solutions alongside robust, real-time aquacultural monitoring technologies.

4.2.2. Policy Development for IoT Adoption

Policy frameworks significantly influence the adoption and effective implementation of IoT technologies. Agricultural IoT adoption is typically encouraged through targeted financial incentives designed to promote precision agrarian practices aimed at conserving critical resources. Empirical evidence indicates that government-supported IoT smart irrigation programs have effectively reduced water usage by approximately 25–50% compared to traditional methods [51,60]. Similarly, precision nutrient management policies have achieved notable improvements in fertilizer use efficiency, reducing waste by up to 40% and minimizing environmental pollution [60].

Conversely, IoT adoption within aquaculture is predominantly driven by regulatory compliance frameworks, particularly stringent environmental monitoring standards related to water quality. Policies mandating continuous IoT-based monitoring of critical parameters, such as ammonia, nitrates, and dissolved oxygen, have significantly accelerated technology uptake, especially for RASs, with documented reductions in nutrient discharge reaching approximately 70% [221,225]. Thus, regulatory compliance serves as a robust mechanism for promoting IoT adoption, contrasting with incentive-driven approaches that are common within agriculture. Nevertheless, substantial policy gaps persist, notably regarding standardization and interoperability. The absence of universally accepted IoT communication standards significantly complicates system integration, reducing adoption incentives for producers, particularly smaller operators [60]. Additionally, unclear data governance frameworks surrounding data ownership, privacy, and sharing inhibit stakeholder collaboration and data exchange, further deterring technology implementation [222].

Infrastructure deficits in rural agricultural areas and remote aquaculture locations further exacerbate adoption barriers. Effective IoT system deployment fundamentally depends on stable connectivity, yet rural regions frequently suffer inadequate broadband infrastructure, undermining real-time data transmission and limiting technological benefits [224]. Addressing these connectivity challenges through dedicated infrastructure investment constitutes a crucial policy priority to facilitate comprehensive IoT adoption.

Finally, inadequate certification and standardization processes for IoT equipment create uncertainty regarding expected performance, reliability, and cybersecurity, complicating procurement and investment decisions. Clear certification guidelines specifying hardware durability, accuracy, reliability, and cybersecurity standards would significantly reduce operational uncertainty, bolstering adoption confidence and facilitating informed decisions [60]. The comparative evaluation suggests agricultural policies effectively leverage direct financial incentives linked explicitly to resource efficiency, while aquaculture policies successfully drive adoption through robust regulatory mandates. Integrating these policy mechanisms by combining financial incentives with clearly defined regulatory standards would likely optimize IoT adoption, enhancing productivity and environmental sustainability across sectors.

4.2.3. Concluding Remarks on Future Developments and Strategic Applications

Further development and broader adoption of agri-AIoT and aqua-AIoT systems represent transformative opportunities to significantly enhance decision-making accuracy, resource efficiency, and environmental sustainability in agriculture and aquaculture. Realizing these benefits, however, requires systematically addressing the current technological,

economic, and policy-related barriers discussed earlier. Future research and strategic actions should explicitly focus on these integrated solutions, prioritizing technological innovation in AI algorithms, robust and standardized sensor hardware, and supportive policy interventions.

Future technological developments might focus explicitly on lightweight, edge-optimized AI algorithms, robust and standardized sensor technologies, and unbiased predictive analytics supported by diverse datasets. Simultaneously, policy interventions emphasizing standardized IoT communication protocols, robust data governance frameworks, improved rural infrastructure, targeted financial incentives, and clear certification standards are essential for accelerating comprehensive technology adoption.

Strategically addressing distinct operational contexts and sector-specific challenges within agriculture and aquaculture through tailored technology and policy solutions provides the most effective pathway toward achieving global sustainable, productive, and resilient food production systems. Continued collaborative efforts across technological innovation, supportive policy frameworks, and interdisciplinary stakeholder engagement will be pivotal in fully realizing the transformative benefits of AI-enhanced IoT systems for sustainable agricultural and aquacultural futures.

5. Conclusions

This review highlights the critical role of IoT-based systems in enhancing sustainability and productivity across agricultural and aquacultural domains. Empirical evidence consistently supports the effectiveness of sensor-driven interventions in improving water and nutrient use efficiency, stabilizing yields, and reducing input waste. In aquaculture, automated monitoring and feeding systems have been particularly effective in improving feed utilization and environmental control under variable conditions. Furthermore, this study was guided by three central research questions:

- (1) What quantifiable benefits do smart sensing and IoT systems offer in agriculture and aquaculture?
- (2) What key technical, economic, and policy-related barriers constrain the adoption of these systems?
- (3) What strategies can support broader and more sustainable implementation?

With respect to the first question, substantial quantitative evidence confirms that IoT deployments lead to significant efficiency and productivity gains, such as 30–50% reductions in water usage, 20–40% improvements in nitrogen utilization, and yield increases exceeding 20% in both sectors. These benefits underscore the transformative potential of smart sensing technologies in resource-constrained food production systems. While the economic feasibility of these systems varies by operational scale, current evidence suggests that basic IoT technologies are increasingly viable for smallholders with targeted support, while more advanced systems are typically adopted by larger commercial farms and facilities.

In addressing the second question, this review identifies persistent limitations, including sensor calibration drift, high upfront investment costs, poor connectivity in rural areas, and insufficient regulatory clarity on data governance. These issues collectively impair adoption, particularly among small- and medium-scale producers. Sensor performance degradation due to environmental factors and biofouling remains a critical concern, as do technical integration challenges and interoperability gaps across platforms.

To answer the third question, this review outlines several strategic directions to overcome the current limitations. These include the development of robust and standardized sensor hardware, the application of AI to enable predictive analytics and automated control (agri-AIoT and aqua-AIoT), and the use of edge computing to reduce latency in

remote deployments. In addition, supportive policy frameworks, including rural infrastructure investments, targeted subsidies, and certification standards, are essential to facilitate scalable adoption.

Underscoring the findings discussed above, future research should prioritize several areas to advance the responsible and inclusive development of IoT technologies in agriculture and aquaculture. First, empirical studies should investigate scale-specific adoption thresholds and economic breakpoints across diverse production and geographic contexts. Second, emerging business models such as “IoT-as-a-service” require further analysis to assess their viability in supporting smallholder inclusion and long-term affordability. Third, challenges related to data integrity, cybersecurity threats, and the ethical use of personal information demand interdisciplinary exploration that spans technical, legal, and governance perspectives. Additionally, efforts should focus on the design of modular and user-friendly systems tailored to the needs of users in low-infrastructure environments, especially where digital literacy or connectivity remains limited. By addressing these cross-cutting issues, future research can help elevate smart sensing technologies from promising innovations to resilient, adaptable foundations for sustainable food systems worldwide. Overall, while smart sensing and AIoT systems offer demonstrated benefits, their widespread application requires coordinated technological innovation, policy support, and operational adaptation. With proper investment and governance, these technologies hold the potential to transform traditional management practices into intelligent, resilient, and sustainable food production systems capable of responding to global environmental and economic uncertainties.

Author Contributions: Conceptualization, L.L. and H.-W.K.; methodology, L.L. and H.-W.K.; validation, L.L. and H.-W.K.; formal analysis, L.L. and H.-W.K.; resources, L.L. and H.-W.K.; writing—original draft preparation, L.L.; writing—review and editing, W.C. and H.-W.K.; supervision, L.L. and H.-W.K.; funding acquisition, L.L. All authors have read and agreed to the published version of the manuscript.

Funding: This work is partially supported by the National Science and Technology Council, Taiwan (MOST 110-2313-B-020-011; NSTC 113-2221-E-020-009-MY3) and Ministry of Agriculture, Taiwan (114 ASm-17.1.5-AD-01).

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Fahad, S.; Bajwa, A.A.; Nazir, U.; Anjum, S.A.; Farooq, A.; Zohaib, A.; Sadia, S.; Nasim, W.; Adkins, S.W.; Saud, S.; et al. Crop Production Under Drought and Heat Stress: Plant Responses and Management Options. *Front. Plant Sci.* **2017**, *8*, 1147. [\[CrossRef\]](#)
2. Ray, D.K.; West, P.; Clark, M.; Gerber, J.; Prishchepov, A.V.; Chatterjee, S. Climate Change Has Likely Already Affected Global Food Production. *PLoS ONE* **2019**, *14*, e0217148. [\[CrossRef\]](#) [\[PubMed\]](#)
3. Jatoi, G.M.; Rahu, M.A.; Karim, S.; Ali, S.M.; Sohu, N. Water Quality Monitoring in Agriculture: Applications, Challenges and Future Prospectus with IoT and Machine Learning. *J. Appl. Eng. Technol.* **2023**, *7*, 46–54. [\[CrossRef\]](#)
4. Zhu, M.; Shang, J. Remote Monitoring and Management System of Intelligent Agriculture Under the Internet of Things and Deep Learning. *Wirel. Commun. Mob. Comput.* **2022**, *2022*, 1206677. [\[CrossRef\]](#)
5. Rosalia, A.C.T.; Mulyaningsih, T. Climate Change Impact on Food Security: A Review. *Sustinere J. Environ. Sustain.* **2023**, *6*, 227–238. [\[CrossRef\]](#)
6. Zaw, A.K.; Charoenratana, S. Climate Change and Food Security at Household Level in the Central Dry Zone in Myanmar. *Manag. Environ. Qual. Int. J.* **2023**, *34*, 1446–1460. [\[CrossRef\]](#)
7. Ali, B.; Zakeri, A.; Llieva, A.; Iliev, O. Reshaping of the Future Farming: From Industry 4. *Am. J. Appl. Sci. Res.* **2023**, *9*, 62–71. [\[CrossRef\]](#)
8. Rifqi, M.; Rofiq, R.M.; Rahman, A.; Rizna Ayu, W.; Nazar, F. Low Carbon Emission Shrimp Farming Development Model. *J. Indones. Sustain. Dev. Plan.* **2022**, *3*, 192–203. [\[CrossRef\]](#)
9. Myers, S.S.; Smith, M.R.; Guth, S.; Golden, C.D.; Vaitla, B.; Mueller, N.D.; Dangour, A.D.; Huybers, P. Climate Change and Global Food Systems: Potential Impacts on Food Security and Undernutrition. *Annu. Rev. Public Health* **2017**, *38*, 259–277. [\[CrossRef\]](#)
10. Rani, P.; Reddy, R.G. Climate Change and Its Impact on Food Security. *Int. J. Environ. Clim. Change* **2023**, *13*, 104–108. [\[CrossRef\]](#)

11. Ali, H.; Menza, M.; Hagos, F.; Hailelassie, A. Impact of Climate-Smart Agriculture Adoption on Food Security and Multidimensional Poverty of Rural Farm Households in the Central Rift Valley of Ethiopia. *Agric. Food Secur.* **2023**, *11*, 62. [\[CrossRef\]](#)
12. Badolo, M. *The Badolo FoodResilience Scientific Framework for Advancing Food Security Resilience to Climate Change in Sub-Saharan Africa*; Center for Open Science: Charlottesville, VA, USA, 2024. [\[CrossRef\]](#)
13. Ngcamu, B.S.; Chari, F. Drought Influences on Food Insecurity in Africa: A Systematic Literature Review. *Int. J. Environ. Res. Public Health* **2020**, *17*, 5897. [\[CrossRef\]](#) [\[PubMed\]](#)
14. Schnitter, R.; Berry, P. The Climate Change, Food Security and Human Health Nexus in Canada: A Framework to Protect Population Health. *Int. J. Environ. Res. Public Health* **2019**, *16*, 2531. [\[CrossRef\]](#) [\[PubMed\]](#)
15. Carpenter, S.R.; Caraco, N.F.; Correll, D.L.; Howarth, R.W.; Sharpley, A.N.; Smith, V.H. Nonpoint pollution of surface waters with phosphorus and nitrogen. *Ecol. Appl.* **1998**, *8*, 559–568. [\[CrossRef\]](#)
16. Guo, J.H.; Liu, X.J.; Zhang, Y.; Shen, J.L.; Han, W.X.; Zhang, W.F.; Christie, P.; Goulding, K.W.T.; Vitousek, P.M.; Zhang, F.S. Significant Acidification in Major Chinese Croplands. *Science* **2010**, *327*, 1008–1010. [\[CrossRef\]](#)
17. Gleiser, M.; Moro, S. Implementation of an IoT-Based Water Quality Monitoring System for Aquaculture. *Int. J. Res. Publ. Rev.* **2023**, *4*, 1449–1452. [\[CrossRef\]](#)
18. Stojanović, N.; Chaudhary, S. Real-Time Water Quality Monitoring in Aquaculture Using IoT Sensors and Cloud-Based Analytics. *Res. J. Comput. Syst. Eng.* **2023**, *4*, 174–187. [\[CrossRef\]](#)
19. Florestiyanto, M.Y.; Ashrianto, P.D.; Yuwono, B.; Himawan, H. Evaluation of Usage Behaviour of IOT-Based Aquaculture Technologies. In *Proceeding on Political and Social Science (PSS)*; RSF Press: Bandung, Indonesia, 2020; pp. 248–256.
20. Prapti, D.R.; Shariff, A.R.M.; Man, H.C.; Ramli, N.M.; Perumal, T.; Shariff, M. Internet of Things (IoT)-based Aquaculture: An Overview of IoT Application on Water Quality Monitoring. *Rev. Aquac.* **2021**, *14*, 979–992. [\[CrossRef\]](#)
21. Anyadike, C.C.; Mbajorgu, C.C.; Ajah, G.N. Review of Aquacultural Production System Models. *Niger. J. Technol.* **2016**, *35*, 448. [\[CrossRef\]](#)
22. Chiu, M.C.; Yan, W.M.; Bhat, S.A.; Huang, N.-F. Development of Smart Aquaculture Farm Management System Using IoT and AI-based Surrogate Models. *J. Agric. Food Res.* **2022**, *9*, 100357. [\[CrossRef\]](#)
23. Kim, J.; Park, E.; Cho, S.; Kwon, K.-W.; Ko, Y.G. Probabilistic Modeling of Fish Growth in Smart Aquaculture Systems. *Ksii Trans. Internet Inf. Syst.* **2023**, *17*, 2259–2277. [\[CrossRef\]](#)
24. Mustafa, S.; Shaleh, S.R.M.; Shapawi, R.; Estim, A.; Ching, F.F.; Ibrahim, A.A.A.; Tuzan, A.D.; Lim, L.S.; Chen, C.-A.; Jimat, A.; et al. Application of Fourth Industrial Revolution Technologies to Marine Aquaculture for Future Food: Imperatives, Challenges and Prospects. *Sustain. Mar. Struct.* **2021**, *3*, 22–31. [\[CrossRef\]](#)
25. Chen, Y.; Zhen, Z.; Yu, H.; Xu, J. Application of Fault Tree Analysis and Fuzzy Neural Networks to Fault Diagnosis in the Internet of Things (IoT) for Aquaculture. *Sensors* **2017**, *17*, 153. [\[CrossRef\]](#)
26. Parra, L.; Sendra, S.; Lloret, J.; Rodrigues, J.J. Design and Deployment of a Smart System for Data Gathering in Aquaculture Tanks Using Wireless Sensor Networks. *Int. J. Commun. Syst.* **2017**, *30*, e3335. [\[CrossRef\]](#)
27. Saha, S.; Rajib, R.H.; Kabir, S. IoT Based Automated Fish Farm Aquaculture Monitoring System. In *Proceedings of the 2018 International Conference on Innovations in Science, Engineering and Technology (ICISSET)*, Chittagong, Bangladesh, 27–28 October 2018; pp. 201–206.
28. Misra, N.N.; Dixit, Y.; Al-Mallahi, A.; Bhullar, M.; Upadhyay, R.; Martynenko, A. IoT, Big Data, and Artificial Intelligence in Agriculture and Food Industry. *IEEE Internet Things J.* **2022**, *9*, 6305–6324. [\[CrossRef\]](#)
29. Valencia-Arias, A.; Dávila, J.R.; Londoño-Celis, W.; Palacios-Moya, L.; Hernández, J.L.; Agudelo-Ceballos, E.; Uribe-Bedoya, H. Research Trends in the Use of the Internet of Things in Sustainability Practices: A Systematic Review. *Sustainability* **2024**, *16*, 2663. [\[CrossRef\]](#)
30. Liu, L.-W.; Hsieh, S.-H.; Lin, S.-J.; Wang, Y.-M.; Lin, W.-S. Rice Blast (*Magnaporthe oryzae*) Occurrence Prediction and the Key Factor Sensitivity Analysis by Machine Learning. *Agronomy* **2021**, *11*, 771. [\[CrossRef\]](#)
31. Wang, X.; Liu, L.; Zhang, W.; Ma, X. Prediction of plant uptake and translocation of engineered metallic nanoparticles by machine learning. *Environ. Sci. Technol.* **2021**, *55*, 7491–7500. [\[CrossRef\]](#)
32. Kuo, H.-W. Tyramine beta hydroxylase-mediated octopamine synthesis pathway in *Litopenaeus vannamei* under thermal, salinity, and *Vibrio alginolyticus* infection stress. *Fish Shellfish. Immunol.* **2023**, *142*, 109096. [\[CrossRef\]](#)
33. Abbasi, R.; Martinez, P.; Ahmad, R. An Ontology Model to Represent Aquaponics 4.0 System's Knowledge. *Inf. Process. Agric.* **2022**, *9*, 514–532. [\[CrossRef\]](#)
34. Chuyễn, T.Đ.; Dien, N.D.; Cuong, N.C.; Thong, V.V. Design and Manufacture Control System for Water Quality Based on IoT Technology for Aquaculture in the Vietnam. *Bull. Electr. Eng. Inform.* **2023**, *12*, 1893–1900. [\[CrossRef\]](#)
35. Muthmainnah, M.; Nashirudin, W.; Sasmitaninghidayah, N.; Chamidah, A.; Mulyono, I. The Development of an IoT-based Automated Temperature and pH Monitoring System to Enhance the Management of Gourami Fish Ponds. *Arpn J. Eng. Appl. Sci.* **2024**, *19*, 294–300. [\[CrossRef\]](#)

36. Liu, L.-W.; Ismail, M.H.; Wang, Y.-M.; Lin, W.-S. Internet of Things based Smart Irrigation Control System for Paddy Field. *AGRIVITA J. Agric. Sci.* **2021**, *43*, 378–389. [[CrossRef](#)]
37. Cavazza, A.; Mas, F.D.; Paoloni, P.; Manzo, M. Artificial Intelligence and New Business Models in Agriculture: A structured Literature Review and future Research Agenda. *Br. Food J.* **2023**, *125*, 436–461. [[CrossRef](#)]
38. Elbaşı, E.; Mostafa, N.; Al-Arnaout, Z.; Zreikat, A.I.; Cina, E.; Varghese, G.; Shdefat, A.Y.; Topcu, A.E.; Abdelbaki, W.; Mathew, S.; et al. Artificial Intelligence Technology in the Agricultural Sector: A Systematic Literature Review. *IEEE Access* **2023**, *11*, 171–202. [[CrossRef](#)]
39. Nagaraja, G.S.; Soppimath, A.B.; Soumya, T.; Abhinith, A. IoT Based Smart Agriculture Management System. In Proceedings of the 2019 4th International Conference on Computational Systems and Information Technology for Sustainable Solution (CSITSS), Bengaluru, India, 20–21 December 2019; pp. 1–5.
40. Akhter, F.; Siddiquei, H.R.; Alahi, M.E.E.; Jayasundera, K.P.; Mukhopadhyay, S.C. An IoT-Enabled Portable Water Quality Monitoring System With MWCNT/PDMS Multifunctional Sensor for Agricultural Applications. *IEEE Internet Things J.* **2022**, *9*, 14307–14316. [[CrossRef](#)]
41. Lal, J.; Vaishnav, A.; Verma, D.K.; Jana, A.; Jayaswal, R.; Chakraborty, A.; Kumar, S.; Devati; Pavankalyan, M. Sahil Emerging Innovations in Aquaculture: Navigating Towards Sustainable Solutions. *Int. J. Environ. Clim. Change* **2024**, *14*, 83–96. [[CrossRef](#)]
42. Pathmudi, V.R.; Khatri, N.; Kumar, S.; Abdul-Qawy, A.S.H.; Vyas, A.K. A Systematic Review of IoT Technologies and Their Constituents for Smart and Sustainable Agriculture Applications. *Sci. Afr.* **2023**, *19*, e01577. [[CrossRef](#)]
43. Sadiku, M.N.O.; Ashaolu, T.J.; Ajayi-Majebi, A.; Musa, S.M. Internet of Things in Agriculture: A Primer. *Int. J. Sci. Adv.* **2021**, *2*, 215–220. [[CrossRef](#)]
44. Hassan, A.; Asif, R.M.; Rehman, A.U.; Nishtar, Z.; Kaabar, M.K.A.; Afsar, K. Design and Development of an Irrigation Mobile Robot. *IAES Int. J. Robot. Autom.* **2021**, *10*, 75. [[CrossRef](#)]
45. Mustapha, U.F.; Alhassan, A.W.; Jiang, D.; Li, G.L. Sustainable Aquaculture Development: A Review on the Roles of Cloud Computing, Internet of Things and Artificial Intelligence (CIA). *Rev. Aquac.* **2021**, *13*, 2076–2091. [[CrossRef](#)]
46. Manikandan, R.; Ranganathan, G.; Bindhu, V. Intelligent Computing and Control Framework for Smart Automated System. *Intell. Autom. Soft Comput.* **2022**, *33*, 173–189. [[CrossRef](#)]
47. de Oliveira, M.P.G.; Zorzeto, T.Q.; Romis, R.d.F.A.; Rodrigues, L.H.A. Can Accuracy Issues of Low-Cost Sensor Measurements Be Overcome with Data Assimilation? *Eng. Agrícola* **2023**, *43*, e20220170. [[CrossRef](#)]
48. Chen, L.-H.; Chen, J.; Chen, C. Effect of Environmental Measurement Uncertainty on Prediction of Evapotranspiration. *Atmosphere* **2018**, *9*, 400. [[CrossRef](#)]
49. Pisanu, T.; Garau, S.; Ortu, P.; Schirru, L.; Macciò, C. Prototype of a Low-Cost Electronic Platform for Real Time Greenhouse Environment Monitoring: An Agriculture 4.0 Perspective. *Electronics* **2020**, *9*, 726. [[CrossRef](#)]
50. Perdinan; Winkler, J.A.; Andresen, J.A. Evaluation of Multiple Approaches to Estimate Daily Solar Radiation for Input to Crop Process Models. *Atmosphere* **2020**, *12*, 8. [[CrossRef](#)]
51. Roberson, T.L.; Badzmierowski, M.J.; Stewart, R.D.; Ervin, E.H.; Askew, S.D.; McCall, D.S. Improving Soil Moisture Assessment of Turfgrass Systems Utilizing Field Radiometry. *Agronomy* **2021**, *11*, 1960. [[CrossRef](#)]
52. Cristo, E.d.; Sgarbossa, J.; Schwerz, F.; Nardini, C.; Tibolla, L.B.; Caron, B.O. Growth and Yield of Soybean Cultivated in Agroforestry Systems. *Rev. Ceres* **2020**, *67*, 165–175. [[CrossRef](#)]
53. Gabriel, C.C.E.; Uribe-Opazo, M.Á.; Dalposso, G.H.; Cima, E.G. Spatial Analysis of Soybean Yield in the Western Mesoregion of Paraná Using Agrometeorological Variables. *Res. Soc. Dev.* **2022**, *11*, e7911325962. [[CrossRef](#)]
54. Munaganuri, R.K.; Rao, Y.N. Cap-DiBiL: An Automated Model for Crop Water Requirement Prediction and Suitable Crop Recommendation in Agriculture. *Environ. Res. Commun.* **2023**, *5*, 095016. [[CrossRef](#)]
55. Oh, Y.-J. Development of Self-generated and LPWA-based Crop Growth Environment Monitoring and Bigdata Analysis System. *Int. J. Adv. Sci. Eng. Inf. Technol.* **2024**, *14*, 1701–1707. [[CrossRef](#)]
56. Nikou, M.; Mavromatis, T. Demonstrating the Use of the Yield-Gap Concept on Crop Model Calibration in Data-Poor Regions: An Application to CERES-Wheat Crop Model in Greece. *Land* **2023**, *12*, 1372. [[CrossRef](#)]
57. Reza, M.N.; Islam, M.N.; Iqbal, M.Z.; Kabir, M.S.N.; Chowdhury, M.; Gulandaz, M.A.; Ali, M.; Jang, M.-K.; Chung, S.O. Spatial, Temporal, and Vertical Variability of Ambient Environmental Conditions in Chinese Solar Greenhouses During Winter. *Appl. Sci.* **2023**, *13*, 9835. [[CrossRef](#)]
58. Prasad, J.; Bhatnagar, V.; Chandra, R. Soil Moisture Sensors for Sustainable Irrigation: Comparison and Calibration. *Int. J. Sustain. Agric. Manag. Inform.* **2019**, *5*, 25. [[CrossRef](#)]
59. Teixeira, J.C.; Santos, R. Exploring the Applicability of Low-Cost Capacitive and Resistive Water Content Sensors on Compacted Soils. *Geotech. Geol. Eng.* **2021**, *39*, 2969–2983. [[CrossRef](#)]
60. Sharma, H.; Shukla, M.K.; Bosland, P.W.; Steiner, R.L. Soil Moisture Sensor Calibration, Actual Evapotranspiration, and Crop Coefficients for Drip Irrigated Greenhouse Chile Peppers. *Agric. Water Manag.* **2017**, *179*, 81–91. [[CrossRef](#)]

61. Abdullah, E.T.; Ibrahim, O.A. Capacitance and Resistivity Measurements of Polythiophene/Metallic Nanoparticles-Based Humidity Sensors. *Iraqi J. Sci.* **2021**, *62*, 1158–1163. [\[CrossRef\]](#)
62. Esmaili, P.; Cavedo, F.; Norgia, M. Liquid Level Sensor Based on Phase-Shifting of Radio-Frequency Wave. *IEEE Sens. J.* **2022**, *22*, 11144–11152. [\[CrossRef\]](#)
63. Nolz, R.; Loiskandl, W. Evaluating Soil Water Content Data Monitored at Different Locations in a Vineyard with Regard to Irrigation Control. *Soil Water Res.* **2017**, *12*, 152–160. [\[CrossRef\]](#)
64. Shintake, J.; Nagai, T.; Ogishima, K. Sensitivity Improvement of Highly Stretchable Capacitive Strain Sensors by Hierarchical Auxetic Structures. *Front. Robot. AI* **2019**, *6*, 127. [\[CrossRef\]](#)
65. Demori, M.; Baù, M.; Dalola, S.; Ferrari, M.; Ferrari, V. Low-Frequency RFID Signal and Power Transfer Circuitry for Capacitive and Resistive Mixed Sensor Array. *Electronics* **2019**, *8*, 675. [\[CrossRef\]](#)
66. Han, S.; Kim, W.J.; Lee, H.J.; Joyce, R.; Lee, J. Continuous and Real-Time Measurement of Plant Water Potential Using an AAO-Based Capacitive Humidity Sensor for Irrigation Control. *Acs Appl. Electron. Mater.* **2022**, *4*, 5922–5932. [\[CrossRef\]](#)
67. Arya, C. Iot Based Precision Farming and Agriculture—Aspects and Technologies. *Math. Stat. Eng. Appl.* **2021**, *70*, 1426–1433. [\[CrossRef\]](#)
68. Kopawar, N.A.; Wankhede, K.G. Internet of Things in Agriculture: A Review. *Int. J. Sci. Res. Sci. Eng. Technol.* **2024**, *11*, 161–165. [\[CrossRef\]](#)
69. Shafi, U.; Mumtaz, R.; García-Nieto, J.; Hassan, S.A.; Zaidi, S.A.R.; Iqbal, N. Precision Agriculture Techniques and Practices: From Considerations to Applications. *Sensors* **2019**, *19*, 3796. [\[CrossRef\]](#) [\[PubMed\]](#)
70. Gupta, D.; Wadhwa, S.; Rani, S.; Khan, Z.; Boulila, W. EEDC: An Energy Efficient Data Communication Scheme Based on New Routing Approach in Wireless Sensor Networks for Future IoT Applications. *Sensors* **2023**, *23*, 8839. [\[CrossRef\]](#)
71. Jagatheesan, M.; Janaki, G. Weather Monitoring System Using IoT for Smart Farming. *ECS Trans.* **2022**, *107*, 17439–17445. [\[CrossRef\]](#)
72. Obaid, M.K.; Abood, B.S.; Alazzai, W.K.; Jasim, L. From Field to Fork: The Role of AI and IoT in Agriculture. *E3S Web Conf.* **2024**, *491*, 02006. [\[CrossRef\]](#)
73. Purnama, S.; Sejati, W. Internet of Things, Big Data, and Artificial Intelligence in the Food and Agriculture Sector. *Int. Trans. Artif. Intell.* **2023**, *1*, 156–174. [\[CrossRef\]](#)
74. Jadhav, N.; Rajnivas, B.; Subapriya, V.; Sivaramakrishnan, S.; Premalatha, S. Enhancing Crop Growth Efficiency Through IoT-enabled Smart Farming System. *EAI Endorsed Trans. Internet Things* **2023**, *10*, 1. [\[CrossRef\]](#)
75. Kosmas, I. Applying Internet of Things in Healthcare: A Survey. *Percept. Reprod. Med.* **2021**, *4*, 25–34. [\[CrossRef\]](#)
76. Kanimozhi, A.; Vadivel, R. Optimized Water Management for Precision Agriculture Using IoT-based Smart Irrigation System. *World J. Adv. Res. Rev.* **2024**, *21*, 802–811. [\[CrossRef\]](#)
77. Raju, K.L.; Vijayaraghavan, V. A Self-Powered, Real-Time, NRF24L01 IoT-Based Cloud-Enabled Service for Smart Agriculture Decision-Making System. *Wirel. Pers. Commun.* **2022**, *124*, 207–236. [\[CrossRef\]](#)
78. Bauer, J.; Aschenbruck, N. Towards a Low-Cost RSSI-based Crop Monitoring. *ACM Trans. Internet Things* **2020**, *1*, 1–26. [\[CrossRef\]](#)
79. Siddharam; Aiswarya, L.; Kambale, J.B.; Rajesh, G. Application of Internet of Things (IoT) in Protected Cultivation: A Review. *Int. J. Environ. Clim. Change* **2023**, *13*, 1518–1529. [\[CrossRef\]](#)
80. Gade, S.; Deshmukh, P.; Amodkar, T.; Dhawale, M.; Jathar, Y. Smart Crop Advisor System Using Iot and Machine Learning. *Int. Res. J. Mod. Eng. Technol. Sci.* **2023**, *5*, 2675–2679. [\[CrossRef\]](#)
81. Singh, P.; Dixit, S.; Sammanit, D.; Krishnan, P. The Automated Farmlands of Tomorrow: An IoT Integration with Farmlands. *IOP Conf. Ser. Mater. Sci. Eng.* **2022**, *1218*, 012048. [\[CrossRef\]](#)
82. Tsiropoulos, Z.; Skoubris, E.; Fountas, S.; Gravalos, I.; Gemtos, T. Development of an Energy Efficient and Fully Autonomous Low-Cost IoT System for Irrigation Scheduling in Water-Scarce Areas Using Different Water Sources. *Agriculture* **2022**, *12*, 1044. [\[CrossRef\]](#)
83. Bouhachlaf, L.; Benslimane, O.; Hajjaji, S.E. Monitoring Soil Elements for Irrigation Management Using Internet of Things (IoT) Sensors. *World Water Policy* **2023**, *9*, 756–766. [\[CrossRef\]](#)
84. Getahun, S.; Kefale, H.; Gelaye, Y. Application of Precision Agriculture Technologies for Sustainable Crop Production and Environmental Sustainability: A Systematic Review. *Sci. World J.* **2024**, *2024*, 2126734. [\[CrossRef\]](#)
85. Rahman, M.B.; Chakma, J.D.; Momin, M.A.; Islam, S.; Uddin, M.A.; Islam, M.A.; Aryal, S. Smart Crop Cultivation System Using Automated Agriculture Monitoring Environment in the Context of Bangladesh Agriculture. *Sensors* **2023**, *23*, 8472. [\[CrossRef\]](#) [\[PubMed\]](#)
86. Ghadge, S.V. The Smart Agriculture System Using IOT and ML. *J. Electr. Syst.* **2024**, *20*, 1539–1548. [\[CrossRef\]](#)
87. Seid, A.M.; Lu, J.; Abishu, H.N.; Ayall, T.A. Blockchain-Enabled Task Offloading with Energy Harvesting in Multi-Uav-Assisted IoT Networks: A Multi-Agent DRL Approach. *IEEE J. Sel. Areas Commun.* **2022**, *40*, 3517–3532. [\[CrossRef\]](#)
88. Lima, M.C.F.; Maria Elisa Damascena de Almeida, L.; Ubierna, C.V.; Coronel, L.C.P.; Bazzo, C.O.G. Automatic Detection and Monitoring of Insect Pests—A Review. *Agriculture* **2020**, *10*, 161. [\[CrossRef\]](#)

89. Xiao, Q.; Zheng, W.; He, Y.; Chen, Z.; Meng, F.; Wu, L. Research on the Agricultural Pest Identification Mechanism Based on an Intelligent Algorithm. *Agriculture* **2023**, *13*, 1878. [\[CrossRef\]](#)
90. Cardoso, B.; Silva, C.; Costa, J.; Ribeiro, B. Internet of Things Meets Computer Vision to Make an Intelligent Pest Monitoring Network. *Appl. Sci.* **2022**, *12*, 9397. [\[CrossRef\]](#)
91. Potamitis, I.; Eliopoulos, P.A.; Rigakis, I. Automated Remote Insect Surveillance at a Global Scale and the Internet of Things. *Robotics* **2017**, *6*, 19. [\[CrossRef\]](#)
92. Ahmed, S.; Marwat, S.N.K.; Brahim, G.B.; Khan, W.U.; Khan, S.N.; Al-Fuqaha, A.; Koziel, S. IoT Based Intelligent Pest Management System for Precision Agriculture. *Sci. Rep.* **2024**, *14*, 31917. [\[CrossRef\]](#) [\[PubMed\]](#)
93. Dong, Y.; Miller, S.R.; Kelley, L. Performance Evaluation of Soil Moisture Sensors in Coarse- And Fine-Textured Michigan Agricultural Soils. *Agriculture* **2020**, *10*, 598. [\[CrossRef\]](#)
94. Deka, N.; Neogi, D.; Islam, A.U.; Borkakoty, S. An Automated Crop Monitoring and Irrigation System with Predictive Analysis. *Periódico Tchê Química* **2024**, *21*, 24–41. [\[CrossRef\]](#)
95. Jiao, J.; Gu, L.; Wang, C.; Wang, Q.; Gao, Y.; Gu, R. Internet of Things_Based Real_Time Farmland Environment Monitoring. *Adv. J. Food Sci. Technol.* **2016**, *11*, 643–650. [\[CrossRef\]](#)
96. Sott, M.K.; Nascimento, L.d.S.; Foguesatto, C.R.; Furstenau, L.B.; Faccin, K.; Zawislak, P.A.; Garcia, B.R.M.; Kong, J.D.; Bragazzi, N.L. A Bibliometric Network Analysis of Recent Publications on Digital Agriculture to Depict Strategic Themes and Evolution Structure. *Sensors* **2021**, *21*, 7889. [\[CrossRef\]](#)
97. Lin, C.; Hu, F.; Peng, J.; Wang, J.; Zhai, R. Segmentation and Stratification Methods of Field Maize Terrestrial LiDAR Point Cloud. *Agriculture* **2022**, *12*, 1450. [\[CrossRef\]](#)
98. Nan, Y.; Zhang, H.; Zheng, J.; Yang, K.; Ge, Y. Low-Volume Precision Spray for Plant Pest Control Using Profile Variable Rate Spraying and Ultrasonic Detection. *Front. Plant Sci.* **2023**, *13*, 1042769. [\[CrossRef\]](#)
99. Gokeda, V.; Yalavarthi, R. Deep Hybrid Model for Pest Detection: IoT-UAV-Based Smart Agriculture System. *J. Phytopathol.* **2024**, *172*, e13381. [\[CrossRef\]](#)
100. Sun, L.; Sun, H.; Cao, N.; Han, X.; Cao, G.; Huo, W.; Zhu, D.; Higgs, R. Intelligent Agriculture Technology Based on Internet of Things. *Intell. Autom. Soft Comput.* **2022**, *32*, 429–439. [\[CrossRef\]](#)
101. Lin, J.Y.; Tsai, H.-L.; Lyu, W.-H. An Integrated Wireless Multi-Sensor System for Monitoring the Water Quality of Aquaculture. *Sensors* **2021**, *21*, 8179. [\[CrossRef\]](#)
102. Engle, C.R.; Senten, J.v. Resilience of Communities and Sustainable Aquaculture: Governance and Regulatory Effects. *Fishes* **2022**, *7*, 268. [\[CrossRef\]](#)
103. Islam, M.J.; Kunzmann, A.; Slater, M.J. Responses of Aquaculture Fish to Climate Change-induced Extreme Temperatures: A Review. *J. World Aquac. Soc.* **2021**, *53*, 314–366. [\[CrossRef\]](#)
104. Kolman, R.; Khudiyi, O.; Kushniryk, O.; Khuda, L.; Prusińska, M.; Wiszniewski, G. Influence of temperature and *Artemia* enriched with ω -3 PUFAs on the early ontogenesis of Atlantic sturgeon, *Acipenser oxyrinchus* Mitchill, 1815. *Aquac. Res.* **2018**, *49*, 1740–1751. [\[CrossRef\]](#)
105. Reverter, M.; Tapissier-Bontemps, N.; Sasal, P.; Saulnier, D. Use of Medicinal Plants in Aquaculture. In *Diagnosis and Control of Diseases of Fish and Shellfish*; Woo, P.T.K., Ed.; Wiley-Blackwell: Hoboken, NJ, USA, 2017; pp. 223–261. [\[CrossRef\]](#)
106. Gentry, R.R.; Lester, S.E.; Kappel, C.V.; White, C.; Bell, T.W.; Stevens, J.; Gaines, S.D. Offshore Aquaculture: Spatial Planning Principles for Sustainable Development. *Ecol. Evol.* **2016**, *7*, 733–743. [\[CrossRef\]](#) [\[PubMed\]](#)
107. Wu, T.-H.; Chen, C.H.; Mao, N.; Lu, S.-T. Fishmeal Supplier Evaluation and Selection for Aquaculture Enterprise Sustainability With a Fuzzy McDM Approach. *Symmetry* **2017**, *9*, 286. [\[CrossRef\]](#)
108. Is-Haak, J.; Koydon, S.; Iampichai, Y.; Ngamphongsai, C. Effects of Ammonia, Temperature and Their Interaction on Oxygen Consumption Rate of Asian Seabass (*Lates calcarifer*) Juveniles. *Agric. Nat. Resour.* **2022**, *56*, 917–924. [\[CrossRef\]](#)
109. Cai, J.; Leung, P. Popularity and parity assessment for more inclusive and balanced aquaculture development. *Sci. Rep.* **2024**, *14*, 25317. [\[CrossRef\]](#) [\[PubMed\]](#)
110. Yang, Z.; Li, D.; Song, J.; Bao, E.; Wang, Q.; Qiu, Y.; Wu, Z. Numerical Calculation and Experimental Analysis of Thermal Environment in Industrialized Aquaculture Facilities. *PLoS ONE* **2023**, *18*, e0290449. [\[CrossRef\]](#)
111. Falconer, L.; Ytteborg, E.; Goris, N.; Lauvset, S.K.; Sandø, A.B.; Hjøllø, S.S. Context Matters When Using Climate Model Projections for Aquaculture. *Front. Mar. Sci.* **2023**, *10*, 1198451. [\[CrossRef\]](#)
112. Tamim, A.T.; Begum, H.; Shachcho, S.A.; Khan, M.M.; Yeboah-Akouwah, B.; Masud, M.; Al-Amri, J.F. Development of IoT Based Fish Monitoring System for Aquaculture. *Intell. Autom. Soft Comput.* **2022**, *32*, 55–71. [\[CrossRef\]](#)
113. Kumar, G.; Hegde, S.; Senten, J.v.; Engle, C.R.; Boldt, N.C.; Parker, M.; Quagrainie, K.K.; Posadas, B.C.; Asche, F.; Dey, M.M.; et al. Economic Contribution of U.S. Aquaculture Farms. *J. World Aquac. Soc.* **2024**, *55*, e13091. [\[CrossRef\]](#)
114. Zhou, J.; Cai, Q.; Zhou, K. The effect of circulation rate on the water quality of container-type recirculation aquaculture system under different feeding rates of *Schizothorax wangchiachii*. *bioRxiv* **2023**. [\[CrossRef\]](#)

115. Akhter, F.; Siddiquei, H.R.; Alahi, M.E.E.; Mukhopadhyay, S.C. Recent Advancement of the Sensors for Monitoring the Water Quality Parameters in Smart Fisheries Farming. *Computers* **2021**, *10*, 26. [\[CrossRef\]](#)
116. Dhinakaran, D. IoT-Based Environmental Control System for Fish Farms with Sensor Integration and Machine Learning Decision Support. *Int. J. Recent Innov. Trends Comput. Commun.* **2023**, *11*, 203–217. [\[CrossRef\]](#)
117. Lemos, C.H.P.; Chung, S.; Caio Paiva Vaz Sampaio, R.; Copatti, C.E. Growth and Biochemical Variables in Amazon Catfish (*Pseudoplatystoma reticulatum* ♀X *Leiarius marmoratus* ♂) Under Different Water pH. *An. Acad. Bras. Ciências* **2018**, *90*, 3573–3581. [\[CrossRef\]](#) [\[PubMed\]](#)
118. Li, H.; Zhang, J.; Ge, X.; Chen, S.; Ma, Z. The Effects of Short-Term Exposure to pH Reduction on the Behavioral and Physiological Parameters of Juvenile Black Rockfish (*Sebastes schlegelii*). *Biology* **2023**, *12*, 876. [\[CrossRef\]](#) [\[PubMed\]](#)
119. Kustija, J.; Andika, F. Control—Monitoring System of Oxygen Level, Ph, Temperature and Feeding in Pond Based on Iot. *Reka Elkomika J. Pengabd. Kpd. Masy.* **2021**, *2*, 1–10. [\[CrossRef\]](#)
120. Wibisono, A.B.; Jayadi, R. Experimental IoT System to Maintain Water Quality in Catfish Pond. *Int. J. Adv. Comput. Sci. Appl.* **2024**, *15*, 393. [\[CrossRef\]](#)
121. Zhao, Y.; Qin, H.; Xu, L.; Yu, H.; Chen, Y. A Review of Deep Learning-Based Stereo Vision Techniques for Phenotype Feature and Behavioral Analysis of Fish in Aquaculture. *Artif. Intell. Rev.* **2024**, *58*, 7. [\[CrossRef\]](#)
122. Sun, Z.; Cai, C.; Guo, F.; Ye, C.; Luo, Y.; Ye, S.; Luo, J.; Zhu, F.; Jiang, C. Oxygen Sensitive Polymeric Nanocapsules for Optical Dissolved Oxygen Sensors. *Nanotechnology* **2018**, *29*, 145704. [\[CrossRef\]](#)
123. Xia, P.; Zhou, H.; Sun, H.; Sun, Q.-F.; Griffiths, R. Research on a Fiber Optic Oxygen Sensor Based on All-Phase Fast Fourier Transform (apFFT) Phase Detection. *Sensors* **2022**, *22*, 6753. [\[CrossRef\]](#)
124. Shehata, N.; Kandas, I.; Samir, E. In-Situ Gold–Ceria Nanoparticles: Superior Optical Fluorescence Quenching Sensor for Dissolved Oxygen. *Nanomaterials* **2020**, *10*, 314. [\[CrossRef\]](#)
125. Li, F.; Wei, Y.; Chen, Y.; Li, D.; Zhang, X. An Intelligent Optical Dissolved Oxygen Measurement Method Based on a Fluorescent Quenching Mechanism. *Sensors* **2015**, *15*, 30913–30926. [\[CrossRef\]](#)
126. Yuan, X.; Wu, X.; He, M.; Lai, J.P.; Sun, H. A Ratiometric Fiber Optic Sensor Based on CdTe QDs Functionalized with Glutathione and Mercaptopropionic Acid for on-Site Monitoring of Antibiotic Ciprofloxacin in Aquaculture Water. *Nanomaterials* **2022**, *12*, 829. [\[CrossRef\]](#) [\[PubMed\]](#)
127. Su, X.; Sutarlie, L.; Loh, X.J. Sensors, Biosensors, and Analytical Technologies for Aquaculture Water Quality. *Research* **2020**, *2020*, 8272705. [\[CrossRef\]](#) [\[PubMed\]](#)
128. Butinyac, M.G.; Montañó, V.A.; Downes, J.; Ruane, N.M.; Ryder, E.; Egan, F.; Staessen, T.; Paull, B.; Murray, E. Continuous Nitrite and Nitrate Monitoring of Recirculating Aquaculture Systems Using a Deployable Ion Chromatography-Based Analyser. *Aquac. Int.* **2023**, *32*, 1013–1026. [\[CrossRef\]](#)
129. Turcios, A.E.; Papenbrock, J. Sustainable Treatment of Aquaculture Effluents—What Can We Learn from the Past for the Future? *Sustainability* **2014**, *6*, 836–856. [\[CrossRef\]](#)
130. Paul, D.; Hall, S. Biochar and Zeolite as Alternative Biofilter Media for Denitrification of Aquaculture Effluents. *Water* **2021**, *13*, 2703. [\[CrossRef\]](#)
131. Rajeev, M.; Jung, I.; Kang, I.; Cho, J.C. Genome-Centric Metagenomics Provides Insights into the Core Microbial Community and Functional Profiles of Biofloc Aquaculture. *Msystems* **2024**, *9*, e0078224. [\[CrossRef\]](#)
132. Ibrahim, H.; Yin, S.; Satyanarayana, M.; Zhu, Y.; Castellano, M.J.; Dong, L. In Planta Nitrate Sensor Using a Photosensitive Epoxy Bioresin. *ACS Appl. Mater. Interfaces* **2022**, *14*, 25949–25961. [\[CrossRef\]](#)
133. Deng, M.; Dai, Z.; Song, K.; Wang, Y.; He, X. Integrating Microbial Protein Production and Harvest Systems into Pilot-Scale Recirculating Aquaculture Systems for Sustainable Resource Recovery: Linking Nitrogen Recovery to Microbial Communities. *Environ. Sci. Technol.* **2021**, *55*, 16735–16746. [\[CrossRef\]](#)
134. Kokkuar, N.; Li, L.; Srisapoom, P.; Dong, S.; Tian, X. Application of Biodegradable Polymers as Carbon Sources in Ex Situ Biofloc Systems: Water Quality and Shift of Microbial Community. *Aquac. Res.* **2021**, *52*, 3570–3579. [\[CrossRef\]](#)
135. Rustini, H.A.; Sapei, A.; Riani, E.; Machfud; Sunaryani, A.; Santoso, A.B.; Nomosatryo, S.; Setiawan, F.; Rahmadya, A. Suitability Assessment of a Volcanic Endorheic Lake for Aquaculture. *IOP Conf. Ser. Earth Environ. Sci.* **2024**, *1359*, 012121. [\[CrossRef\]](#)
136. Gil-Núñez, J.C.; Martínez-Córdova, L.R.; Servín-Villegas, R.; Magallón-Barajas, F.J.; Bórquez-López, R.A.; González-Galavíz, J.R.; Casillas-Hernández, R. Production of *Penaeus vannamei* in Low Salinity, Using Diets Formulated with Different Protein Sources and Percentages. *Lat. Am. J. Aquat. Res.* **2020**, *48*, 396–405. [\[CrossRef\]](#)
137. Pattusamy, A.; Hittinahalli, C.M.; Chadha, N.K.; Sawant, P.B.; Krishna, H.; Verma, A.K. Water Budgeting for Culture of *Penaeus vannamei* (Boone, 1931) in Earthen Grow-out Ponds Using Inland Saline Groundwater. *Aquac. Res.* **2022**, *53*, 4521–4530. [\[CrossRef\]](#)
138. Madrigal, I.E.V.; Valenzuela-Quinónez, W.; Esparza-Leal, H.M.; Quiroz, G.R.; Noriega, E.A.A. Efecto De La Composición Iónica Sobre El Crecimiento Y La Supervivencia De Camarón Blanco *Litopenaeus vannamei* Cultivado en Agua De Pozo De Baja Salinidad. *Rev. Biol. Mar. Oceanogr.* **2019**, *52*, 103–112. [\[CrossRef\]](#)

139. Rao, C.V.; Chari, N.V.H.K.; Muralikrishna, R. The Impact of Shrimp Pond Effluent on Water Quality of Vasishta Godavari Estuary With Respect to Brackishwater Aquaculture, East Coast of India. *Egypt. J. Aquat. Biol. Fish.* **2019**, *23*, 245–255. [\[CrossRef\]](#)
140. Santos, H.M.; Tsai, C.Y.; Maquiling, K.R.A.; Tayo, L.L.; Mariatulqabtiah, A.R.; Lee, C.-W.; Chuang, K.P. Diagnosis and Potential Treatments for Acute Hepatopancreatic Necrosis Disease (AHPND): A Review. *Aquac. Int.* **2019**, *28*, 169–185. [\[CrossRef\]](#)
141. Wardhany, V.A.; Yuliandoko, H.; Subono, A.M.U.H.; Astawa, I.G.P. Smart System and Monitoring of *Vanammei* Shrimp Ponds. *Int. J. Adv. Sci. Eng. Inf. Technol.* **2021**, *11*, 1366. [\[CrossRef\]](#)
142. Hou, D.; Zhou, R.; Zeng, S.; Wei, D.; Deng, X.; Xing, C.; Yu, L.; Deng, Z.; Wang, H.; Weng, S.; et al. Intestine Bacterial Community Composition of Shrimp Varies Under Low- And High-Salinity Culture Conditions. *Front. Microbiol.* **2020**, *11*, 589164. [\[CrossRef\]](#)
143. Stiller, K.T.; Kolarevic, J.; Lazado, C.C.; Gerwins, J.; Good, C.; Summerfelt, S.T.; Mota, V.C.; Espmark, Å.M.O. The Effects of Ozone on Atlantic Salmon Post-Smolt in Brackish Water—Establishing Welfare Indicators and Thresholds. *Int. J. Mol. Sci.* **2020**, *21*, 5109. [\[CrossRef\]](#) [\[PubMed\]](#)
144. Putri, F.R.; Budiyo, I.R.; Afaulu, R.A.P. Is Water Oxidation-Reduction Potential (ORP) Value Relevant for Aquaculture Applications in Shrimp Farming? *E3S Web Conf.* **2023**, *442*, 02015. [\[CrossRef\]](#)
145. Halla, P.T.H.B.; Lalel, H.; Santoso, P. Short Communication: Comparison of the Water Environment Aspects and Production of Nile Tilapia (*Oreochromis niloticus*) Between Biofloc and Conventional Aquaculture Systems in Tropical Dryland Region. *Int. J. Trop. Drylands* **2023**, *7*, 102. [\[CrossRef\]](#)
146. Siskandar, R.; Wiyoto, W.; Hendriana, A.; Ekasari, J.; Kusumah, B.R.; Halim, G.; Nugraha, I.J. Automated Redox Monitoring System (ARMS): An Instrument for Measuring Dissolved Oxygen Levels Using a Potential Redox Sensor (ORP) in a Prototype of Shrimp Farming Pond with an Internet-Based Monitoring System. *J. Aquac. Fish Health* **2022**, *11*, 238–246. [\[CrossRef\]](#)
147. Qu, X.; Xia, W.; Wang, R.; Zhang, Y.; Xie, Z.; Trushenski, J.; Chen, Y. Effects of Aquaculture on Lakes in the Central Yangtze River Basin, China, II: Benthic Macroinvertebrates. *North Am. J. Aquac.* **2018**, *80*, 369–378. [\[CrossRef\]](#)
148. Tolon, M.T.; Tokaç, A.; Kostak, E.N.; Strehse, C. A Photonic Sensor System for Real-Time Monitoring of Turbidity Changes in Aquaculture. *North Am. J. Aquac.* **2024**, *86*, 424–432. [\[CrossRef\]](#)
149. Amijar, M.I.S.M.; Ramli, N.M.; Nurulhuda, K.; Aziz, S.A. Portable Spectrophotometer for Water Quality Monitoring in Recirculating Aquaculture Systems. *IOP Conf. Ser. Earth Environ. Sci.* **2024**, *1359*, 012028. [\[CrossRef\]](#)
150. Ende, S.S.; Larceva, E.; Bögnér, M.; Lugert, V.; Slater, M.J.; Henjes, J. Low Turbidity in Recirculating Aquaculture Systems (RAS) Reduces Feeding Behavior and Increases Stress-Related Physiological Parameters in Pikeperch (*Sander lucioperca*) During Grow-Out. *Transl. Anim. Sci.* **2021**, *5*, txab223. [\[CrossRef\]](#)
151. Munif, M.M.; Nasir, H.J.A.; Ahmad, M.I. Optimizing Ant Colony System Algorithm with Rule-Based Data Classification for Smart Aquaculture. *Indones. J. Electr. Eng. Comput. Sci.* **2024**, *33*, 261. [\[CrossRef\]](#)
152. Mota, V.C.; Striberny, A.; Verstege, G.C.; Difford, G.F.; Lazado, C.C. Evaluation of a Recirculating Aquaculture System Research Facility Designed to Address Current Knowledge Needs in Atlantic Salmon Production. *Front. Anim. Sci.* **2022**, *3*, 876504. [\[CrossRef\]](#)
153. Jayaraj, K.; Saravanan, P.; Bhowmick, G.D. Performance Evaluation of Aquaponics-Waste-Based Biochar as a Cathode Catalyst in Sediment Microbial Fuel Cells for Integrated Multitrophic Aquaculture Systems. *Energies* **2023**, *16*, 5922. [\[CrossRef\]](#)
154. Mahmud, H.; Rahaman, M.A.; Hazra, S.; Ahmed, S. IoT Based Integrated System to Monitor the Ideal Environment for Shrimp Cultivation With Android Mobile Application. *Eur. J. Inf. Technol. Comput. Sci.* **2023**, *3*, 22–27. [\[CrossRef\]](#)
155. Alahmad, T.; Neményi, M.; Nyéki, A. Applying IoT Sensors and Big Data to Improve Precision Crop Production: A Review. *Agronomy* **2023**, *13*, 2603. [\[CrossRef\]](#)
156. Vatin, N.; Joshi, S.K.; Acharya, P.; Sharma, R.; Rajasekhar, N. Precision Agriculture and Sustainable Yields: Insights from IoT-Driven Farming and the Precision Agriculture Test. *Bio Web Conf.* **2024**, *86*, 01091. [\[CrossRef\]](#)
157. Hussein, A.H.A.; Jabbar, K.A.; Mohammed, A.; Jasim, L. Harvesting the Future: AI and IoT in Agriculture. *E3S Web Conf.* **2024**, *477*, 00090. [\[CrossRef\]](#)
158. Senoo, E.E.K.; Anggraini, L.; Kumi, J.A.; Karolina, L.B.; Akansah, E.; Sulyman, H.A.; Mendonça, I.; Aritsugi, M. IoT Solutions with Artificial Intelligence Technologies for Precision Agriculture: Definitions, Applications, Challenges, and Opportunities. *Electronics* **2024**, *13*, 1894. [\[CrossRef\]](#)
159. Alturif, G.; Saleh, W.; El-Bary, A.A.; Osman, R.A. Towards Efficient IoT Communication for Smart Agriculture: A Deep Learning Framework. *PLoS ONE* **2024**, *19*, e0311601. [\[CrossRef\]](#)
160. Chacko, N.M.; Narendra, V.G.; Balachandra, M.; Rathinam, S. Exploring IoT-Blockchain Integration in Agriculture: An Experimental Study. *IEEE Access* **2023**, *11*, 130439–130450. [\[CrossRef\]](#)
161. Monteleone, S.; Moraes, E.A.d.; Faria, B.T.d.; Aquino, P.T.; Maia, R.F.; Neto, A.T.; Toscano, A. Exploring the Adoption of Precision Agriculture for Irrigation in the Context of Agriculture 4.0: The Key Role of Internet of Things. *Sensors* **2020**, *20*, 7091. [\[CrossRef\]](#)
162. Bakthavatchalam, K.; Karthik, B.; Thiruvengadam, V.S.; Muthal, S.; Jose, D.V.; Kotecha, K.; Vijayakumar, V. IoT Framework for Measurement and Precision Agriculture: Predicting the Crop Using Machine Learning Algorithms. *Technologies* **2022**, *10*, 13. [\[CrossRef\]](#)

163. Quí, V.K.; Nguyen, V.-H.; Anh, D.V.; Quí, N.M.; Ban, N.T.; Lanza, S.; Randazzo, G.; Muzirafuti, A. IoT-Enabled Smart Agriculture: Architecture, Applications, and Challenges. *Appl. Sci.* **2022**, *12*, 3396. [\[CrossRef\]](#)
164. Tsouros, D.C.; Bibi, S.; Sarigiannidis, P. A Review on UAV-Based Applications for Precision Agriculture. *Information* **2019**, *10*, 349. [\[CrossRef\]](#)
165. Alharbi, H.A.; Aldossary, M. Energy-Efficient Edge-Fog-Cloud Architecture for IoT-Based Smart Agriculture Environment. *IEEE Access* **2021**, *9*, 110480–110492. [\[CrossRef\]](#)
166. Wang, T.; Lu, Y.; Cao, Z.; Shu, L.; Zheng, X.; Liu, A.; Xie, M. When Sensor-Cloud Meets Mobile Edge Computing. *Sensors* **2019**, *19*, 5324. [\[CrossRef\]](#) [\[PubMed\]](#)
167. Cocos, H.-N.; Merkl, D. Decentralized Data Processing on the Edge—Accessing Wireless Sensor Networks with Edge Computing. In Proceedings of the 16th International Conference on Applied Computing, Cagliari, Italy, 7–9 November 2019; pp. 265–268.
168. Sepulveda, F.; Thangraj, J.S.; Pulliam, J. The Edge of Exploration: An Edge Storage and Computing Framework for Ambient Noise Seismic Interferometry Using Internet of Things Based Sensor Networks. *Sensors* **2022**, *22*, 3615. [\[CrossRef\]](#) [\[PubMed\]](#)
169. Rajalakshmi, M.; Gunasekaran, K. Effective Nutrient Removal From Aquaculture Wastewater Utilizing an Indoor Nutrient Film Technique Hydroponic System. *Environ. Qual. Manag.* **2024**, *34*, e22262. [\[CrossRef\]](#)
170. Klonoff, D.C. Fog Computing and Edge Computing Architectures for Processing Data from Diabetes Devices Connected to the Medical Internet of Things. *J. Diabetes Sci. Technol.* **2017**, *11*, 647–652. [\[CrossRef\]](#) [\[PubMed\]](#)
171. Sathyamoorthy, S.; Matthew, U.O.; Adekunle, T.S.; Okafor, N. Advances and Challenges in IoT Sensors Data Handling and Processing in Environmental Monitoring Networks. *JSMT* **2024**, *5*, 40–60. [\[CrossRef\]](#)
172. Guastella, D.A.; Marcillaud, G.; Valenti, C. Edge-Based Missing Data Imputation in Large-Scale Environments. *Information* **2021**, *12*, 195. [\[CrossRef\]](#)
173. Ooi, M.P.; Sohail, S.; Huang, V.; Hudson, N.; Baughman, M.; Rana, O.; Hinze, A.; Chard, K.; Chard, R.; Foster, I.; et al. Measurement and Applications: Exploring the Challenges and Opportunities of Hierarchical Federated Learning in Sensor Applications. *IEEE Instrum. Meas. Mag.* **2023**, *26*, 21–31. [\[CrossRef\]](#)
174. Wang, W.; Feng, C.; Zhang, B.; Gao, H. Environmental Monitoring Based on Fog Computing Paradigm and Internet of Things. *IEEE Access* **2019**, *7*, 127154–127165. [\[CrossRef\]](#)
175. Sun, L.; Wang, Z.; Jiang, J.; Kim, Y.; Joo, B.; Zheng, S.; Lee, S.; Yu, W.J.; Kong, B.S.; Yang, H. In-Sensor Reservoir Computing for Language Learning via Two-Dimensional Memristors. *Sci. Adv.* **2021**, *7*, eabg1455. [\[CrossRef\]](#)
176. Xu, Y.P.; Jin, J.; Zeng, S.; Zhang, Y.; Xiao, Q. Development and Evaluation of An Iot-Based Portable Water Quality Monitoring System for Aquaculture. *Inmateh Agric. Eng.* **2023**, *70*, 359–368. [\[CrossRef\]](#)
177. Ghazali, D.M.; Nor, N.S.M.; Mohaini, M.A.; Saputra, H.M. Smart IoT Based Monitoring System for Fish Breeding. *J. Adv. Res. Appl. Mech.* **2023**, *104*, 1–11. [\[CrossRef\]](#)
178. Sung, W.T.; Isa, I.G.T.; Hsiao, S.J. An IoT-Based Aquaculture Monitoring System Using Firebase. *Comput. Mater. Contin.* **2023**, *76*, 2179–2200. [\[CrossRef\]](#)
179. Ragavi, R.; Ramya, A.; Sowndharya, G.; Sunmathy, S. Hybrid Power Source in Fish Farming with Help of IoT Technology. *Ir. Interdiscip. J. Sci. Res.* **2024**, *8*, 61–70. [\[CrossRef\]](#)
180. Marini, R.; Mikhaylov, K.; Pasolini, G.; Buratti, C. Low-Power Wide-Area Networks: Comparison of LoRaWAN and NB-IoT Performance. *IEEE Internet Things J.* **2022**, *9*, 21051–21063. [\[CrossRef\]](#)
181. Pointl, M.; Fuchs-Hanusch, D. Assessing the Potential of LPWAN Communication Technologies for Near Real-Time Leak Detection in Water Distribution Systems. *Sensors* **2021**, *21*, 293. [\[CrossRef\]](#) [\[PubMed\]](#)
182. Alqurashi, H.; Bouabdallah, F.; Khairullah, E.F. SCAP SigFox: A Scalable Communication Protocol for Low-Power Wide-Area IoT Networks. *Sensors* **2023**, *23*, 3732. [\[CrossRef\]](#) [\[PubMed\]](#)
183. Singh, R.K.; Rahmani, M.H.; Weyn, M.; Berkvens, R. Joint Communication and Sensing: A Proof of Concept and Datasets for Greenhouse Monitoring Using LoRaWAN. *Sensors* **2022**, *22*, 1326. [\[CrossRef\]](#)
184. Abhijna, K.C.; Prasanna, R.; Ganavi, G.S.; Ghosh, D. Sensors, Internet and Cloud Computing-Based Smart Agriculture. *Int. J. Eng. Res. Comput. Sci. Eng.* **2022**, *9*, 62–65. [\[CrossRef\]](#)
185. Bhaskaran, H.S.; Gordon, M.; Neethirajan, S. Development of a Cloud-Based IoT System for Livestock Health Monitoring Using AWS and Python. *bioRxiv* **2024**. [\[CrossRef\]](#)
186. Pierleoni, P.; Concetti, R.; Belli, A.; Palma, L. Amazon, Google and Microsoft Solutions for IoT: Architectures and a Performance Comparison. *IEEE Access* **2020**, *8*, 5455–5470. [\[CrossRef\]](#)
187. Chakraborty, S.; Aithal, P.S. Let Us Create an IoT Inside the AWS Cloud. *Int. J. Case Stud. Bus. It Educ.* **2023**, *7*, 211–219. [\[CrossRef\]](#)
188. Barros, T.G.F.; Eronides, F.d.S.N.; Neto, J.A.; Andre, G.M.D.S.; Aquino, V.B.; Teixeira, E.S. The Anatomy of IoT Platforms—A Systematic Multivocal Mapping Study. *IEEE Access* **2022**, *10*, 72758–72772. [\[CrossRef\]](#)
189. Mijuskovic, A.; Ullah, I.; Bemthuis, R.; Meratnia, N.; Havinga, P. Comparing Apples and Oranges in IoT Context: A Deep Dive Into Methods for Comparing IoT Platforms. *IEEE Internet Things J.* **2021**, *8*, 1797–1816. [\[CrossRef\]](#)
190. Li, X.; Zou, B. An automated data engineering pipeline for anomaly detection of IoT sensor data. *arXiv* **2021**, arXiv:2109.13828.

191. Dauda, A.; Flauzac, O.; Nolot, F. A Survey on IoT Application Architectures. *Sensors* **2024**, *24*, 5320. [\[CrossRef\]](#)
192. Baeza, V.M.; Marban, M.A. High Altitude Platform Stations Aided Cloud-Computing Solution for Rural-Environment IoT Applications. *Comput. Netw. Commun.* **2023**, *11*, 2107. [\[CrossRef\]](#)
193. El-Basioni, B.M.M.; El-Kader, S.M.A. Laying the Foundations for an IoT Reference Architecture for Agricultural Application Domain. *IEEE Access* **2020**, *8*, 190194–190230. [\[CrossRef\]](#)
194. Liu, C.-S.; Chen, X.-T.; Shih, W.-Y.; Lin, C.C.; Yen, J.-H.; Huang, C.-J.; Yen, Y.-T. Smart Water Quality Monitoring Technology for Fish Farms Using Cellphone Camera Sensor. *Sens. Mater.* **2023**, *35*, 3019. [\[CrossRef\]](#)
195. Vo, T.T.E.; Ko, H.; Huh, J.H.; Kim, Y. Overview of Smart Aquaculture System: Focusing on Applications of Machine Learning and Computer Vision. *Electronics* **2021**, *10*, 2882. [\[CrossRef\]](#)
196. Islam, S.I.; Ahammad, F.; Mohammed, H.H. Cutting-edge Technologies for Detecting and Controlling Fish Diseases: Current Status, Outlook, and Challenges. *J. World Aquac. Soc.* **2024**, *55*, e13051. [\[CrossRef\]](#)
197. Kaur, G.; Adhikari, N.; Krishnapriya, S.; Wawale, S.G.; Malik, R.Q.; Zamani, A.S.; Falcón, J.P.; Osei-Owusu, J. Recent Advancements in Deep Learning Frameworks for Precision Fish Farming Opportunities, Challenges, and Applications. *J. Food Qual.* **2023**, *2023*, 4399512. [\[CrossRef\]](#)
198. Dabas, A. Application of Support Vector Machines in Machine Learning. *TechRxiv* **2024**. [\[CrossRef\]](#)
199. Fang-an, D. Some New Properties of Wd-fuzzy Implication Algebras. *Math. Lett.* **2021**, *7*, 25. [\[CrossRef\]](#)
200. Martins, J.A.; Azevedo, A.M.; Almeida, A.C.d.; da Silva, L.C.R.; Fernandes, A.C.G.; Valadares, N.R.; Aspiazú, I. Fuzzy Logic Is a Powerful Tool for the Automation of Milk Classification. *Acta Sci. Technol.* **2022**, *44*, e57860. [\[CrossRef\]](#)
201. Li, B.; Shahzad, M.; Khan, H.; Bashir, M.; Ullah, A.; Siddique, M.H. Sustainable Smart Agriculture Farming for Cotton Crop: A Fuzzy Logic Rule Based Methodology. *Sustainability* **2023**, *15*, 13874. [\[CrossRef\]](#)
202. Gutierrez, D.V.; Castañeda, J.; Vélez, G.C. Design and Implementation of a Greenhouse for Organic Tomatoes with Fuzzy Logic. *J. Southwest Jiaotong Univ.* **2023**, *58*, 14. [\[CrossRef\]](#)
203. Nguyen, X.T.; Phung, K.P.; Dang, Q.H.; Vo, X.H.; Vū, H.T. A Target Threat Assessment Method for Application in Air Defense Command and Control Systems. *J. Russ. Univ. Radioelectron.* **2023**, *26*, 90–98. [\[CrossRef\]](#)
204. Putra, S.D.; Heriansyah, H.; Cahyadi, E.F.; Anggriani, K.; Jaya, M.H.I.S. Development of Smart Hydroponics System Using AI-based Sensing. *J. Infotel* **2024**, *16*, 474–485. [\[CrossRef\]](#)
205. Silalahi, A.O.; Sinambela, A.; Panggabean, H.M.; Pardosi, J.T.N. Smart Automated Fish Feeding Based on IoT System Using LoRa TTGO SX1276 and Cayenne Platform. *Eureka Phys. Eng.* **2023**, *66*–79. [\[CrossRef\]](#)
206. Tiutiunnyk, H.; Iermakova, O. Implementing Recirculation Systems in Ukrainian Aquaculture: Intersectoral Economic Measures and Proposals. In Proceedings of the V International Scientific and Practical Conference, Cambridge, UK, 18 August 2023; pp. 13–16.
207. Rao, M.P.T.; Anil, B.; Siddhartha, G.; Shyam, Y.; Rao, B.P. Smart Farming Decision Support System for Precision Agriculture. *J. Nonlinear Anal. Optim.* **2024**, *15*, 1751–1758. [\[CrossRef\]](#)
208. Vijayakumar, S.; Nithya, N.; Saravanane, P.; Mariadoss, A.; Subramanian, E. Revolutionizing Rice Farming: Maximizing Yield with Minimal Water to Sustain the Hungry Planet. In *Irrigation Systems and Applications*; Sultan, M., Imran, M., Ahmad, F., Eds.; IntechOpen: Rijeka, Croatia, 2023. [\[CrossRef\]](#)
209. Mary, D.L.; Ramakrishnan, M.; Singh, A. Performance of Smart Farming Through Drip Irrigation and Managing of Fertilizers and Pesticides Through IoT and GSM. *Int. J. Eng. Adv. Technol.* **2020**, *9*, 3746–3750. [\[CrossRef\]](#)
210. Khan, M.A.A.; Ali, A.; Ashraf, I.; Siddiqui, M.T.H.; Knox, J. Evaluating Socio-economic and Environmental Factors Influencing Farm-level Water Scarcity in Punjab, Pakistan*. *Irrig. Drain.* **2020**, *70*, 797–808. [\[CrossRef\]](#)
211. Marazky, M.E. Effect of Smart Irrigation Controllers Units on the Performance and Productivity of Subsurface and Surface Drip Irrigation Systems for Tomato Crop in Arid Regions. *J. Soil Sci. Agric. Eng.* **2015**, *6*, 27–46. [\[CrossRef\]](#)
212. Arouna, A.; Dzomeku, I.K.; Abdul-Ganiyu, S.; Rahman, N.A. Water Management for Sustainable Irrigation in Rice (*Oryza sativa* L.) Production: A Review. *Agronomy* **2023**, *13*, 1522. [\[CrossRef\]](#)
213. Sasikumar, R.; Lincy, L.L.; Sathyan, A.; Chellapandi, P. Design, Development, and Deployment of a Sensor-Based Aquaculture Automation System. *Aquac. Int.* **2024**, *32*, 6431–6447. [\[CrossRef\]](#)
214. Vijayaram, S.; Mahendran, K.; Ringø, E.; Razafindralambo, H.; Kannan, S.; Sun, Y. Biogenic Dietary Promoters in Aquaculture: Nature-Based Solutions for Enhancing Growth, Health, and Sustainability. *Ann. Anim. Sci.* **2024**, *in press*. [\[CrossRef\]](#)
215. Gidiagba, J.O.; Nwaobia, N.K.; Biu, P.W.; Ezeigweneme, C.; Umoh, A.A. Review on the Evolution and Impact of Iot-Driven Predictive Maintenance: Assessing Advancements, Their Role in Enhancing System Longevity, and Sustainable Operations in Both Mechanical and Electrical Realms. *Comput. Sci. It Res. J.* **2024**, *5*, 166–189. [\[CrossRef\]](#)
216. Kimothi, S.; Thapliyal, A.; Singh, R.; Rashid, M.; Gehlot, A.; Akram, S.V.; Javed, A.R. Comprehensive Database Creation for Potential Fish Zones Using IoT and ML With Assimilation of Geospatial Techniques. *Sustainability* **2023**, *15*, 1062. [\[CrossRef\]](#)

217. Muthoka, M.; Ouko, K.O.; Mboya, J.B.; Ndambuki, M.N.; Outa, N.; Ogello, E.; Obiero, K.; Ogola, R.J.O.; Midamba, D.C.; Njogu, L. Socio-economic Impacts of Climate Change and Adaptation Actions Among Smallholder Fish Farmers in Sub-Saharan Africa. *Aquac. Fish Fish.* **2024**, *4*, e182. [\[CrossRef\]](#)
218. Pullo, S.; Pareschi, R.; Piantadosi, V.; Salzano, F.; Carlini, R. Integrating IOTA's Tangle with the Internet of Things for Sustainable Agriculture: A Proof-of-Concept Study on Rice Cultivation. *Informatics* **2023**, *11*, 3. [\[CrossRef\]](#)
219. Wang, Z.; Chen, X.; Shi, L.; Zhang, X.; Hu, Y.J. Productivity Versus Environmental Sustainability: A Broadscale Assessment of Freshwater Aquaculture's Technical Efficiency and Ecological Efficiency in China's Inland Provinces. *J. World Aquac. Soc.* **2024**, *55*, e13057. [\[CrossRef\]](#)
220. Begum, S.S.; Mashl, H.; Karthikeyan, B.; Alanazi, F.Z. A Prediction of Heart Disease Using IoT Based ThingSpeak Basis and Deep Learning Method. *J. Adv. Res. Appl. Sci. Eng. Technol.* **2024**, *47*, 166–179. [\[CrossRef\]](#)
221. Zhao, Y.; Zhang, H.; Jin, Q.; Jia, D.; Liu, T. Ratiometric Optical Fiber Dissolved Oxygen Sensor Based on Fluorescence Quenching Principle. *Sensors* **2022**, *22*, 4811. [\[CrossRef\]](#) [\[PubMed\]](#)
222. Kumar, P.; Jain, K.K.; Munilkumar, S.; Sudhagar, A. Alternate Feeding Strategies for Optimum Nutrient Utilization and Reducing Feed Cost for Semi-Intensive Practices in Aquaculture System-a Review. *Agric. Rev.* **2017**, *38*, 145–151. [\[CrossRef\]](#)
223. Narsale, S.A.; Prakash, P.; Mohale, H.P.; Baraiya, R.; Sheikh, S.; Kirtikumar, P.B.; Mansukhbhai, C.R.; Kadam, R.V.; Tekam, I. Precision Aquaculture: A Way Forward for Sustainable Agriculture. *J. Exp. Agric. Int.* **2024**, *46*, 83–97. [\[CrossRef\]](#)
224. Zuriati, Z.; Supriyatna, A.R.; Arifin, O. Design and Development of Feeding Automation System and Water Quality Monitoring on Freshwater Fish Cultivation. *Iop Conf. Ser. Earth Environ. Sci.* **2022**, *1012*, 012077. [\[CrossRef\]](#)
225. Li, E.; Wang, X.; Chen, K.; Xu, C.; Qin, J.G.; Chen, L. Physiological Change and Nutritional Requirement of Pacific White Shrimp *Litopenaeus vannamei* at Low Salinity. *Rev. Aquac.* **2015**, *9*, 57–75. [\[CrossRef\]](#)
226. Preden, J.; Kaugerand, J.; Suurjaak, E.; Astapov, S.; Motus, L.; Pahtma, R. Data to decision: Pushing situational information needs to the edge of the network. In Proceedings of the 2015 IEEE International Multi-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision, Orlando, FL, USA, 9–12 March 2015; pp. 158–164.
227. Liu, F.; Huang, Z.; Wang, L. Energy-Efficient Collaborative Task Computation Offloading in Cloud-Assisted Edge Computing for IoT Sensors. *Sensors* **2019**, *19*, 1105. [\[CrossRef\]](#)
228. Liu, L.-W.; Lu, C.-T.; Wang, Y.-M.; Lin, K.-H.; Ma, X.; Lin, W.-S. Rice (*Oryza sativa* L.) Growth Modeling Based on Growth Degree Day (GDD) and Artificial Intelligence Algorithms. *Agriculture* **2022**, *12*, 59. [\[CrossRef\]](#)
229. Liu, L.-W.; Ma, X.; Wang, Y.-M.; Lu, C.-T.; Lin, W.-S. Using artificial intelligence algorithms to predict rice (*Oryza sativa* L.) growth rate for precision agriculture. *Comput. Electron. Agric.* **2021**, *187*, 106286. [\[CrossRef\]](#)
230. Liu, L.; Ma, X. Prediction of Soil Field Capacity and Permanent Wilting Point Using Accessible Parameters by Machine Learning. *AgriEngineering* **2024**, *6*, 2592–2611. [\[CrossRef\]](#)
231. Shafi, U.; Mumtaz, R.; Iqbal, N.; Zaidi, S.M.H.; Zaidi, S.A.R.; Hussain, I.; Mahmood, Z. A Multi-Modal Approach for Crop Health Mapping Using Low Altitude Remote Sensing, Internet of Things (IoT) and Machine Learning. *IEEE Access* **2020**, *8*, 112708–112724. [\[CrossRef\]](#)

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.